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Chapter 1

Trigonometric Functions

As mentioned in an earlier chapter, in a right triangle where one angle is θ, the sine of θ is the length of the side opposite θ divided by the length of the hypotenuse.

The sine function is defined for any real number. We treat that real number θ as an angle, we draw a ray from the origin out to the unit circle. The y value of that point is the sine. For example, the $\sin(\frac{4\pi}{3})$ is $-\sqrt{3}/2$ (see figure 1.1).

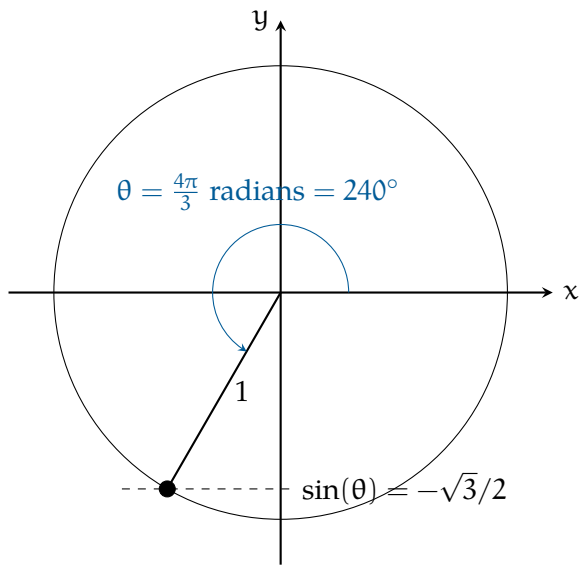


Figure 1.1: $\sin\frac{4\pi}{3} = \frac{-\sqrt{3}}{2}$

(Note that in this section, we will be using radians instead of degrees unless otherwise noted. While degrees are more familiar to most people, engineers and mathematicians nearly always use radians when solving problems. Your calculator should have a radians mode and a degrees mode; you want to be in radians mode.)

Similarly, we define cosine using the unit circle. To find the cosine of θ, we draw a ray from the origin at the angle θ. The x component of the point where the ray intersects the unit circle is the cosine of θ (shown in figure 1.2).

From this description, it is easy to see why $\sin(\theta)^2 + \cos(\theta)^2 = 1$. They are the legs of a right triangle with a hypotenuse of length 1.

It should also be easy to see why $\sin(\theta) = \sin(\theta + 2\pi)$: Each time you go around the circle,

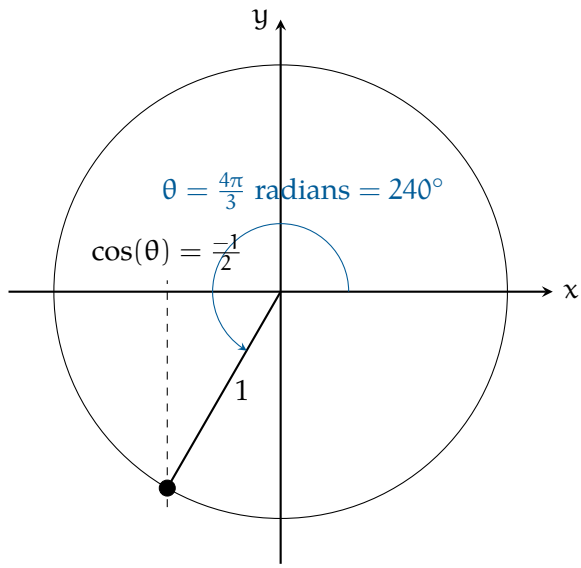


Figure 1.2: $\cos\frac{4\pi}{3} = \frac{-1}{2}$

you come back to where you started.

Can you see why $\cos(\theta) = \sin(\theta + \pi/2)$? Turn the picture sideways.

1.1 Graphs of sine and cosine

Here is a graph of $y = \sin(x)$:

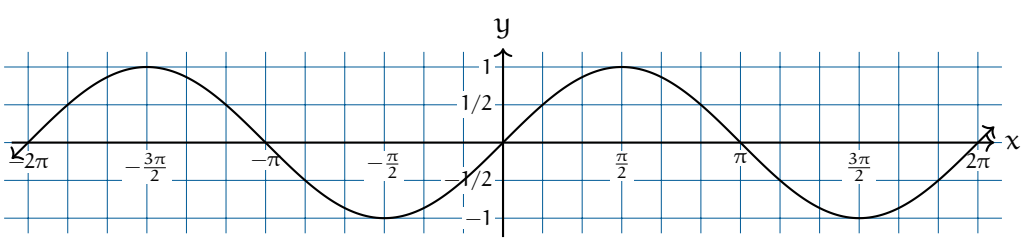


Figure 1.3: $y = \sin(x)$.

It looks like waves, right? It goes forever to the left and right. Remembering that $\cos(\theta) = \sin(\theta + \pi/2)$, we can guess what the graph of $y = \cos(x)$ looks like:

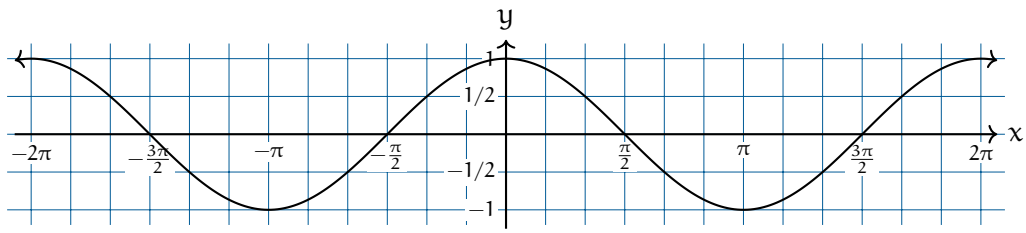


Figure 1.4: $y = \cos(x)$.

1.2 Plot cosine in Python

Create a file called `cos.py`:

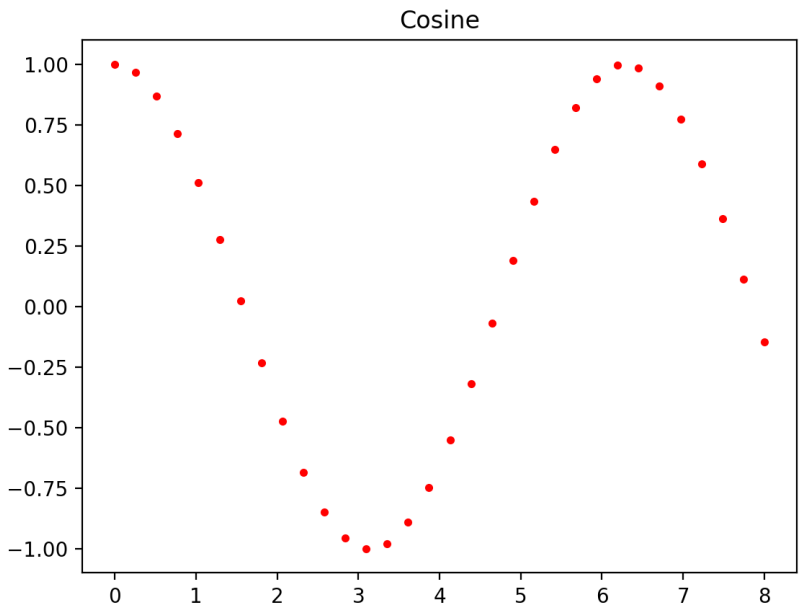
```
import numpy as np
import matplotlib.pyplot as plt

until = 8.0

# Make a plot of cosine
thetas = np.linspace(0, until, 32)
cosines = []
for theta in thetas:
    cosines.append(np.cos(theta))

# Plot the data
fig, ax = plt.subplots()
ax.plot(thetas, cosines, 'r.', label="Cosine")
ax.set_title("Cosine")
plt.show()
```

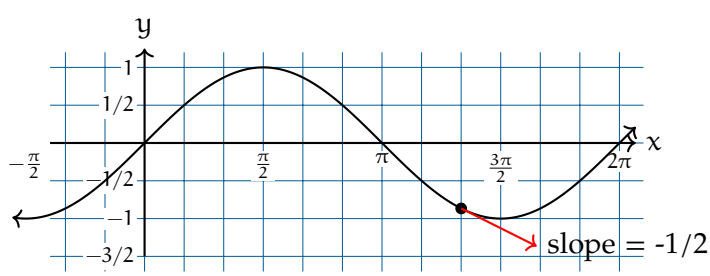
This will plot 32 points on the cosine wave between 0 and 8. When you run it, you should see something like this:



1.3 Derivatives of trigonometric functions

Here is a wonderful property of sine and cosine functions: At any point θ, the slope of the sine graph at θ equals $\cos(\theta)$.

For example, we know that $\sin(4\pi/3) = -(1/2)\sqrt{3}$ and $\cos(4\pi/3) = -1/2$. If we drew a line tangent to the sine curve at this point, it would have a slope of -1/2:



We say "The derivative of the sine function is the cosine function."

Can you guess the derivative of the cosine function? For any θ, the slope of the graph of the $\cos(\theta)$ is $-\sin(\theta)$.

Exercise 1 Derivatives of Trig Functions Practice 1

Use the limit definition of a derivative to show that $\frac{d}{dx}\cos x = -\sin x$

Working Space

Answer on Page 69

The derivatives of all the trigonometric functions are presented below:

$\frac{d}{dx}\sin x = \cos x$	$\frac{d}{dx}\csc x = -\csc x \cdot \cot x$
$\frac{d}{dx}\cos x = -\sin x$	$\frac{d}{dx}\sec x = \sec x \cdot \tan x$
$\frac{d}{dx}\tan x = \sec^2 x$	$\frac{d}{dx}\cot x = -\csc^2 x$

Example: Find the derivative of $f(x)$ if $f(x) = x^2 \sin x$ **Solution**: Using the product rule, we find that:

$$\frac{d}{dx}f(x) = (x^2)\frac{d}{dx}(\sin x) + (\sin x)\frac{d}{dx}(x^2)$$

Taking the derivatives:

$$= x^2(\cos x) + 2x(\sin x)$$

Exercise 2 Derivatives of Trig Functions 2

Find the derivative of the following functions:

Working Space

1. $f(x) = \frac{\sec x}{1+\tan x}$
2. $y = \sec t \tan t$
3. $f(\theta) = \frac{\theta}{4-\tan\theta}$
4. $f(t) = 2\sec t - \csc t$
5. $f(\theta) = \frac{\sin\theta}{1+\cos\theta}$
6. $f(x) = \sin x \cos x$

Answer on Page 70

1.4 A weight on a spring

Let's say you fill a rollerskate with heavy rocks and attach it to the wall with a stiff spring. If you push the skate toward the wall and release it, it will roll back and forth. Engineers would say "The skate will oscillate." Intuitively, you can probably guess:

- If the spring is stronger, the skate will oscillate more times per minute.
- If the rocks are lighter, the skate will oscillate more times per minute.

The force that the spring exerts on the skate is proportional to how far its length is from its relaxed length. When you buy a spring, the manufacturer advertises its "spring rate", which is in pounds per inch or newtons per meter. If a spring has a rate of 5 newtons per meter, that means that if you stretch or compress it 10 cm, it will push back with a force of 0.5 newtons. If you stretch or compress it 20 cm, it will push back with a force of 1 newton.

Let's write a simulation of the skate-on-a-spring. Duplicate `cos.py`, and name the new copy `spring.py`. Add code to implement the simulation:

```
import numpy as np
import matplotlib.pyplot as plt

until = 8.0

# Constants
mass = 100 # kg
spring_constant = -1 # newtons per meter displacement
time_step = 0.01 # s

# Initial state
displacement = 1.0 # height above equilibrium in meters
velocity = 0.0
time = 0.0 # seconds

# Lists to gather data
displacements = []
times = []

# Run it for a little while
while time <= until:
    # Record data
    displacements.append(displacement)
    times.append(time)

    # Calculate the next state
    time += time_step
    displacement += time_step * velocity
    force = spring_constant * displacement
```

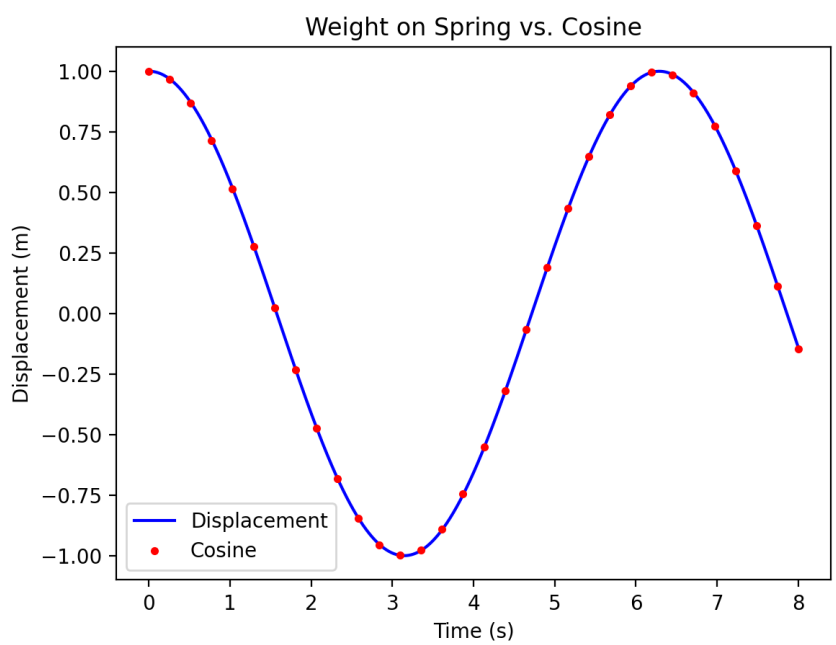
```
    acceleration = force / mass
    velocity += acceleration

# Make a plot of cosine
thetas = np.linspace(0, until, 32)
cosines = []
for theta in thetas:
    cosines.append(np.cos(theta))

# Plot the data
fig, ax = plt.subplots()
ax.plot(times, displacements, 'b', label="Displacement")
ax.plot(thetas, cosines, 'r.', label="Cosine")

ax.set_title("Weight on Spring vs. Cosine")
ax.set_xlabel("Time (s)")
ax.set_ylabel("Displacement (m)")
ax.legend()
plt.show()
```

When you run it, you should get a plot of your spring and the cosine graph on the same plot.



The position of the skate is following a cosine curve. Why?

Because a sine or cosine waves happen whenever the acceleration of an object is proportional to -1 times its displacement. Or in symbols:

$$a \propto -p$$

where a is acceleration and p is the displacement from equilibrium.

Remember that if you take the derivative of the displacement, you get the velocity. And if you take the derivative of that, you get acceleration. So, the weight on the spring must follow a function f such that

$$f(t) \propto -f''(t)$$

Remember that the derivative of the $\sin(\theta)$ is $\cos(\theta)$.

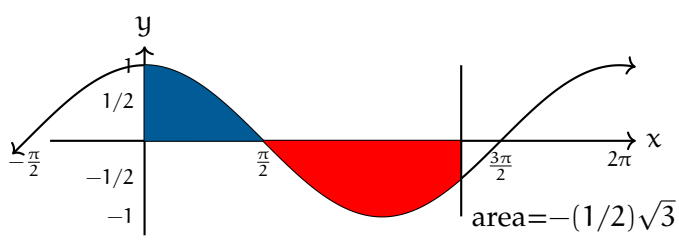
And the derivative of the $\cos(\theta)$ is $-\sin(\theta)$

These sorts of waves have an almost-magical power: Their acceleration is proportional to -1 times their displacement.

Thus, sine waves of various magnitudes and frequencies are ubiquitous in nature and technology.

1.5 Integral of sine and cosine

If we take the area between the graph and the x axis of the cosine function (and if the function is below the x axis, it counts as negative area), from 0 to $4\pi/3$, we find that it is equal to $-(1/2)\sqrt{3}$.



We say "The integral of the cosine function is the sine function."

1.5.1 Integrals of Trig Functions Practice

Exercise 3

Evaluate the following integrals:

1. $\int \sec x \tan x \, dx$

Working Space

Answer on Page 70

CHAPTER 2

Trigonometric Identities

2.1 The Unit Circle

There are some values of $\sin\theta$ and $\cos\theta$ that will be useful to know off the top of your head. The Unit Circle will help you in this memorization process (see figure 2.1). When a circle of radius 1 is centered at the origin, the Cartesian coordinates of any point on the circle correspond to the values of cosine and sine of the angle above the horizontal (how far you've rotated from the positive x-axis).

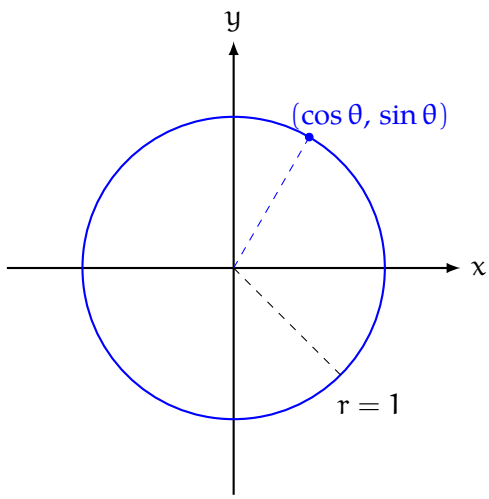


Figure 2.1: The Unit Circle is a circle with radius 1 centered at the origin

Let's take a closer look at a triangle in the first quadrant to see why this is true. Imagine some point on the circle, (x_o, y_o). Drawing a line from that point back to the origin creates an angle θ between the imaginary line and the positive x-axis (see figure 2.2). Extending an imaginary vertical down to $(x_o, 0)$, then an imaginary horizontal from $(x_o, 0)$ to the origin, creates a right triangle. What can we say about the legs of the triangle?

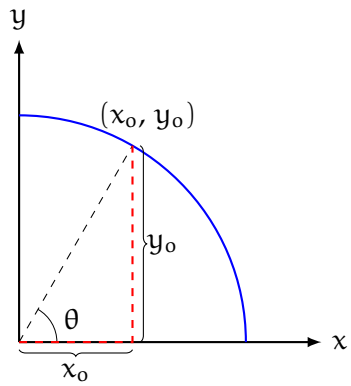


Figure 2.2: Drawing a line from any point on the circle to the origin creates an angle with the horizontal

Recall SOH-CAH-TOA from a previous chapter. This acronym tells us that, for a right triangle, the sine of an angle is given by the ratio of the length of the leg opposite the angle to the hypotenuse. In our case, then, $\sin\theta = \frac{y_o}{1} = y_o$. [Remember: We are dealing with the Unit Circle, which has a radius of one. Examining figure 2.2 shows you that the hypotenuse of the imaginary triangle is the same as the circle's radius.] This means that the y-coordinate of any point on the Unit Circle is the sine of the angle of rotation from the horizontal.

Exercise 4

In a similar manner as we did with $\sin\theta$ above, prove the x-coordinate of any point on the unit circle is equal to $\cos\theta$, where θ is the angle of rotation from the horizontal.

Working Space

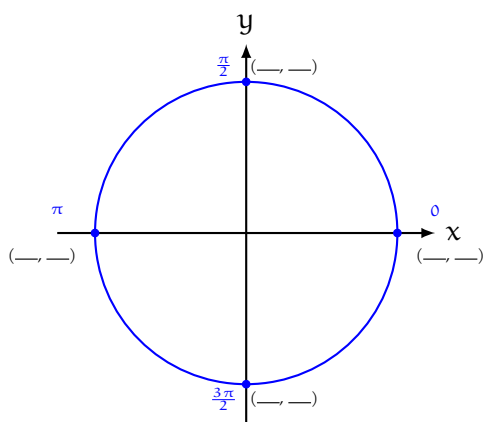
Answer on Page 70

From these exercises, we can see that each (x, y) coordinate on the circle is equal to $(\cos(\theta), \sin(\theta))$.

Exercise 5

Fill in the unit circle with the coordinates for $\theta = 0$, $\pi/2$, π, and $3\pi/2$. Use this to determine:

1. $\sin \frac{\pi}{2}$
2. $\cos \frac{3\pi}{2}$
3. $\sin \pi$
4. $\cos -\pi$



Working Space

Answer on Page 70

2.1.1 Exact Values of Key Angles

We will examine two triangles. First, a 30-60-90 triangle, then a 45-45-90 triangle. As shown in figure 2.3, you can get a 30-60-90 triangle with hypotenuse 1 by dividing an equilateral triangle in half. We will label the horizontal leg of the 30-60-90 triangle A and the vertical leg B.

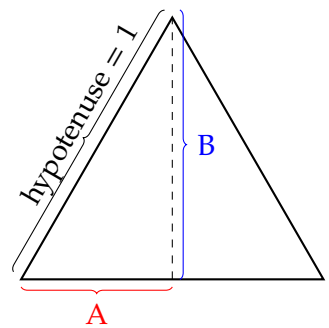


Figure 2.3: A 30-60-90 triangle is made by vertically bisecting an equilateral triangle

From the figure, we see that the length of A is half that of the hypotenuse, which in this case is $\frac{1}{2}$. This means the $\cos 60^\circ = \cos \frac{\pi}{3} = \frac{1}{2}$. To find the length of side B, we can use the Pythagorean theorem:

$$B^2 = C^2 - A^2, \text{ where C is the hypotenuse}$$

$$B^2 = 1^2 - \left(\frac{1}{2}\right)^2$$

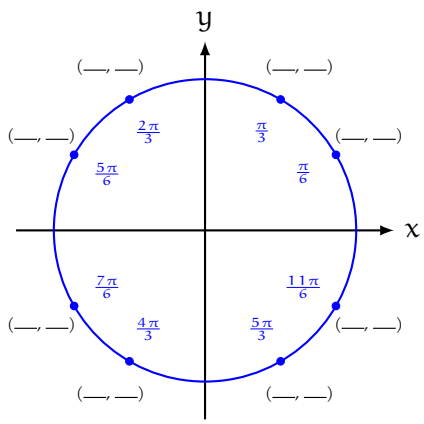
$$B^2 = \frac{3}{4}$$

$$B = \frac{\sqrt{3}}{2}$$

Therefore, $\sin 60^\circ = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$.

Exercise 6

Use symmetry to complete the blank unit circle below. (Hint: We just showed that the (x, y) coordinate for $\frac{\pi}{3}$ is $(1/2, \sqrt{3}/2)$).



Working Space

Answer on Page 71

Now we will look at a 45-45-90 triangle (see figure 2.4), which will allow us to complete our Unit Circle. Recall that a 45-45-90 triangle is an isosceles triangle in addition to being a right triangle. This means both the legs are the same length. Using the Pythagorean theorem, we would say $A = B$. We also know that $C = 1$, since our triangle is inscribed in the unit circle. Let's find A:

$$A^2 + B^2 = C^2$$
$$A^2 + A^2 = 1^2$$
$$2A^2 = 1$$
$$A^2 = \frac{1}{2}$$
$$A = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Therefore, each leg has a length of $\sqrt{2}/2$, and the (x, y) coordinates for $\theta = 45° = \pi/4$ are $(\sqrt{2}/2, \sqrt{2}/2)$.

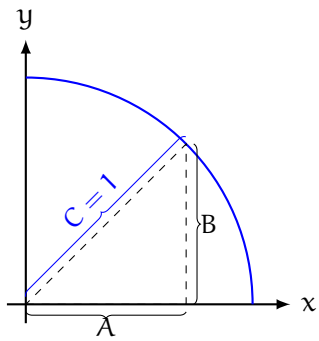
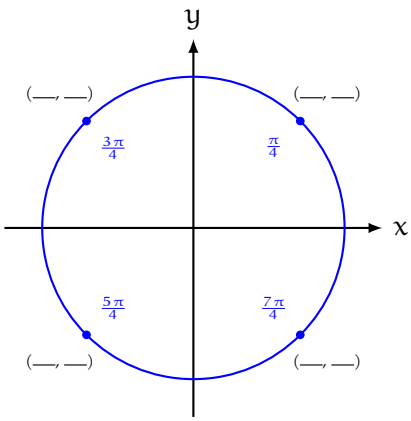


Figure 2.4: The two legs of a 45-45-90 triangle are the same length

Exercise 7

Use symmetry to complete the blank unit circle below.



Working Space

Answer on Page 71

Exercise 8

Without a calculator and using only your completed unit circles, determine the value requested (angles are given in radians unless otherwise indicated).

1. $\cos \frac{3\pi}{2}$
2. $\sin \frac{\pi}{4}$
3. $\sin -\frac{\pi}{6}$
4. $\cos \frac{4\pi}{3}$
5. $\sin \frac{3\pi}{4}$
6. $\cos -\frac{\pi}{3}$
7. $\sin 45°$
8. $\sin 270°$
9. $\sin -60°$
10. $\sin 150°$

Working Space

Answer on Page 72

2.2 Sum and Difference Formulas

Consider 4 points on the unit circle: B at $(1, 0)$, Q at some angle β, P at some angle α, and A at angle $\alpha - \beta$ (see figure 2.5).

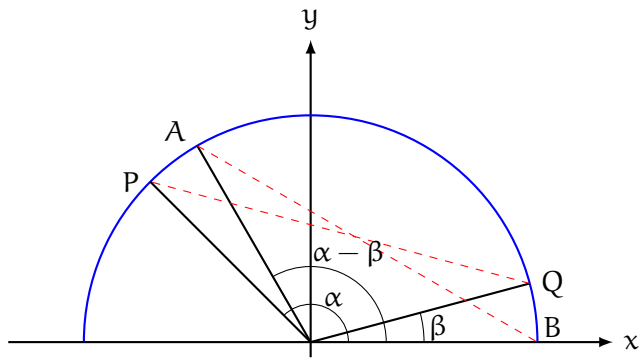


Figure 2.5: $\overline{AB} = \overline{PQ}$

The distance from P to Q is the same as the distance from A to B, since ΔPOQ is a rotation of ΔAOB. Because this is a Unit Circle, $P = (\cos\alpha, \sin\alpha)$, $Q = (\cos\beta, \sin\beta)$, and $A = (\cos\alpha - \beta, \sin\alpha - \beta)$. Let's use the distance formula to find the length of $\overline{PQ}$:

$$\overline{PQ} = \sqrt{(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2} =$$

$$\sqrt{\cos^2\alpha - 2\cos\alpha\cos\beta + \cos^2\beta + \sin^2\alpha - 2\sin\alpha\sin\beta + \sin^2\beta} =$$

$$\sqrt{\left(\cos^2\alpha + \sin^2\alpha\right) + \left(\cos^2\beta + \sin^2\beta\right) - 2\cos\alpha\cos\beta - 2\sin\alpha\sin\beta}$$

Recall that for any angle, θ, $\sin^2\theta + \cos^2\theta = 1$. Substituting this identity, we see that:

$$\overline{PQ} = \sqrt{1 + 1 - 2\sin\alpha\sin\beta - 2\cos\alpha\cos\beta} = \sqrt{2 - 2\sin\alpha\sin\beta - 2\cos\alpha\cos\beta}$$

Let's leave this simplified equation for $\overline{PQ}$ alone for the moment and similarly find $\overline{AB}$:

$$\overline{AB} = \sqrt{[\cos(\alpha - \beta) - 1]^2 + [\sin(\alpha - \beta) - 0]^2} =$$

$$\sqrt{\cos^2(\alpha - \beta) - 2\cos(\alpha - \beta) + 1 + \sin^2(\alpha - \beta)} =$$

$$\sqrt{\cos^2(\alpha - \beta) + \sin^2(\alpha - \beta) + 1 - 2\cos(\alpha - \beta)}$$

$$= \sqrt{2 - 2\cos(\alpha - \beta)} = \overline{AB}$$

Recall that we've established $\overline{AB} = \overline{PQ}$. We can set the statements equal to each other:

$$\sqrt{2 - 2\sin\alpha\sin\beta - 2\cos\alpha\cos\beta} = \sqrt{2 - 2\cos(\alpha - \beta)}$$

Squaring both sides and subtracting 2, we find:

$$-2\sin\alpha\sin\beta - 2\cos\alpha\cos\beta = -2\cos(\alpha-\beta)$$

Finally, we can divide both sides by negative 2 to get the difference of angles formula for cosine:

$$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

There are similar formulas for the sine and cosine of the sum of two angles, and for the sine of the difference of two angles, which we won't derive here.

Sum and Difference Formulas

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$
$$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$
$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$$

Exercise 9

Without a calculator, find the exact value requested:

1. $\sin\frac{\pi}{12}$
2. $\cos\frac{7\pi}{12}$
3. $\tan\frac{13\pi}{12}$ (hint: $\tan\theta = \sin\theta/\cos\theta$)

Working Space

Answer on Page 72

2.3 Double and Half Angle Formulas

We can easily derive a formula for twice an angle by letting $\alpha = \beta$ for a sum formula.

Example: Derive a formula for $\cos 2\theta$ in terms of trigonometric functions of θ.

Solution: Using the sum formula for cosine, we see that:

$$\cos 2\theta = \cos(\theta + \theta)$$

$$= \cos\theta\cos\theta - \sin\theta\sin\theta = \cos^2\theta - \sin^2\theta$$

Noting that $\sin^2\theta = 1 - \cos^2\theta$: $\boxed{\cos 2\theta = 2\cos^2\theta - 1}$ Alternatively, we could note that $\cos^2\theta = 1 - \sin^2\theta$: $\boxed{\cos 2\theta = 1 - 2\sin^2\theta}$

Or additionally, $\boxed{\cos 2\theta = \cos^2\theta - \sin^2\theta}$

Exercise 10

Derive a formula for $\sin 2\theta$ in terms of trigonometric functions of θ.

Working Space

Answer on Page 72

We can use these double-angle formulas for find half-angle formulas. Consider the double-angle formula for cosine:

$$\cos 2\theta = 2\cos^2\theta - 1$$

Let $\theta = \alpha/2$, then:

$$\cos\alpha = 2\cos^2(\alpha/2) - 1$$

Rearranging to solve for $\cos(\alpha/2)$:

$$2\cos^2(\alpha/2) = \cos\alpha + 1$$

$$\cos^2(\alpha/2) = \frac{\cos\alpha + 1}{2}$$

$$\cos(\alpha/2) = \pm\sqrt{\frac{1+\cos\alpha}{2}}$$

Exercise 11

Derive a formula for $\sin(\alpha/2)$.

Working Space

Answer on Page 73

There are two identities that will be very useful for integrals in a future chapter:

Squared Trigonometric Identities

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$

These are just specific re-writings of the half-angle identities.

Chapter 3

Volumes of Common Solids

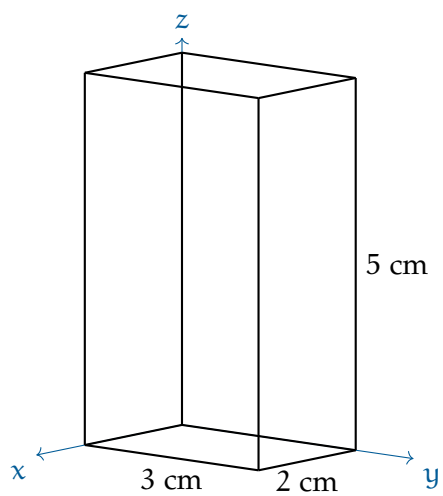
3.1 Rectangular Prism

The volume of a rectangular solid is the product of its three dimensions. If a block of ice is 5 cm tall, 3 cm wide, and 2 cm deep, its volume is $5 \times 3 \times 2 = 30$ cubic centimeters.

Volume of a rectangular solid.

A rectangular solid with height h, width w and length/depth l has volume:

$$V = lwh$$



A cubic centimeter is the same as a milliliter. A milliliter of ice weighs about 0.92 grams. This means the block of ice would have a mass of $30 \times 0.92 = 27.6$ grams.

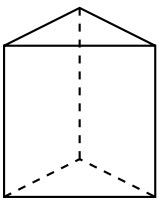
3.2 Triangular Prism

Triangular prisms are 3D versions of triangles (imagine stretching a triangle out of the page). It has 2 triangular faces and 3 rectangular faces.

Volume of a triangular prism.

Recall the area of a triangle is $V = \frac{1}{2}wh$ where w is the width or base and h is the height of the triangle. A triangular prism with height h, width w and length/depth l has volume:

$$V = \frac{1}{2}lwh$$



3.3 Spheres

Volume of a Sphere

A sphere with a radius of r has a volume of

$$v = \frac{4}{3}\pi r^3$$

(For completeness, the surface area of that sphere would be

$$a = 4\pi r^2$$

Note that a circle of radius r is one quarter of this: πr^2.)

Exercise 12 Flying Sphere

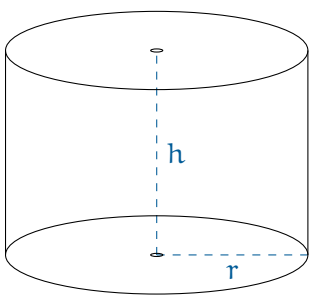
Working Space

An iron sphere is traveling at 5 m/s (and is not spinning). The sphere has a radius of 1.5 m. Iron has a density of 7,800 kg per cubic meter. How much kinetic energy does the sphere have?

Answer on Page 73

3.4 Cylinders

The base and the top of a right cylinder are identical circles. The circles are on parallel planes. The sides are perpendicular to those planes.



Volume of a cylinder

The volume of the right cylinder of radius r and height h is given by:

$$v = \pi r^2 h$$

In other words, it is the area of the base times the height.

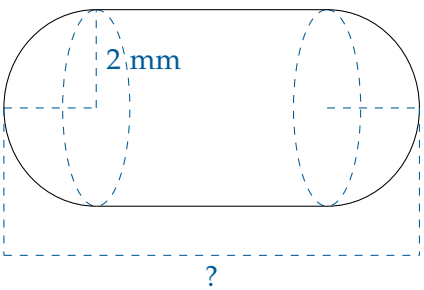
Exercise 13 Tablet

Working Space

A drug company needs to create a tablet with volume of 90 cubic millimeters.

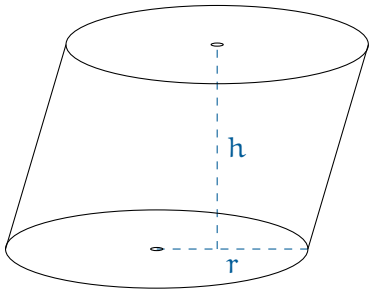
The tablet will be a cylinder with half spheres on each end. The radius will be 2mm.

How long do they need to make the tablet to be?



Answer on Page 73

What if the base and top are identical, but the sides aren't perpendicular to the base? This is called *oblique cylinder*.

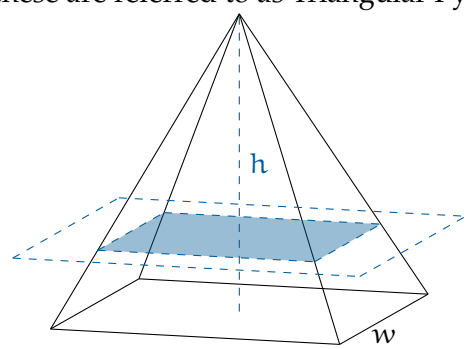


The volume is still the height times the area of the base. Note, however, that the height is measured perpendicular to the bottom and top.

Why is this the case?

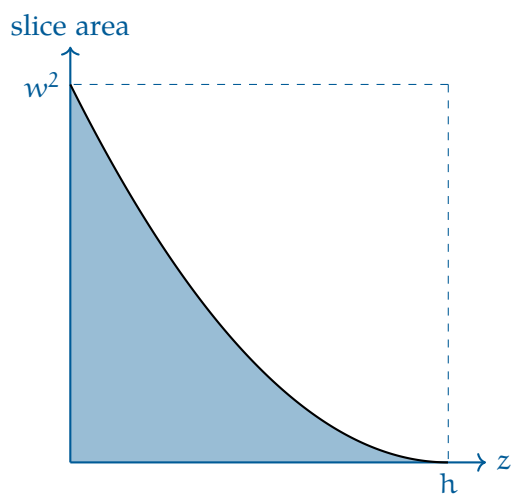
3.5 Pyramids

On a solid with a flat base, the line that we use to measure height is always perpendicular to the plane of the base. We can take slices through the solid that are parallel to that base plane. For example, if we have a pyramid with a square base, each slice will be a square — small squares near the top, larger squares near the bottom. The sides of the pyramids are all triangles, so these are referred to as Triangular Pyramids, just pyramids, or sometimes, Tetrahedrons.



We can figure out the area of the slice at every height z. For example, at $z = 0$, the slice would have area w^2. At $z = h$, the slice would have zero area. What about an arbitrary z in between? The edge of the square would be $w(1 - \frac{z}{h})$. The area of the slice would be $w^2(1 - \frac{z}{h})^2$

The graph of this would look like this:



The volume is given by the area under the curve and above the axis. Once you learn integration, you will be extra good at finding the area under the curve. In this case, we will just tell you that the colored region in the picture is one third of the rectangle.

Thus, the area of a square-based pyramid is $\frac{1}{3}hw^2$.

In fact:

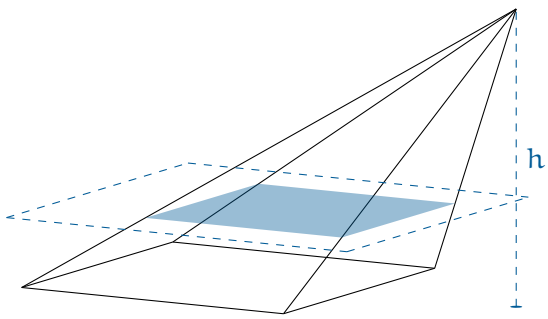
Volume of a pyramid

The volume of pyramid whose base has an area of b and height h is given by:

$$V = \frac{1}{3}hb$$

Regardless of the shape of the base.

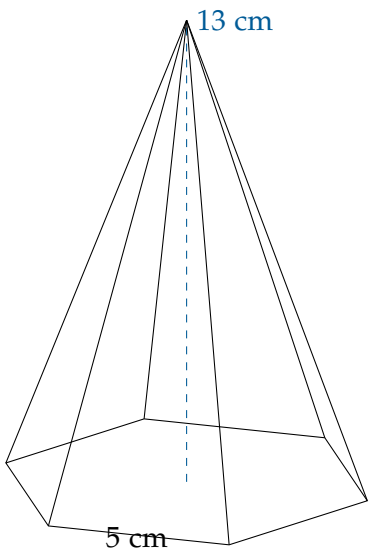
Note that this is true even for oblique pyramids:



Exercise 14 Hexagon-based Pyramid

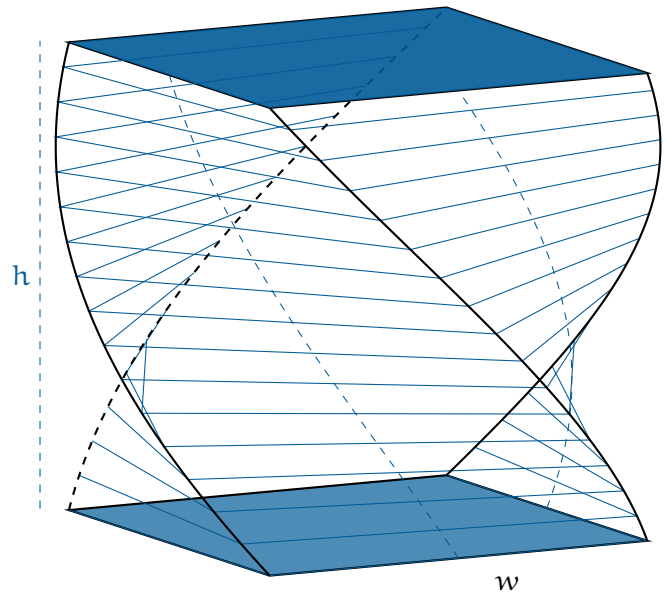
Working Space

There is a pyramid with a regular hexagon for a base. Each edge is 5 cm long. The pyramid is 13 cm tall. What is its volume?



Answer on Page 74

Note that plotting the area of each slice and finding the area under the curve will let you find the area of many things. For example, let's say that you have a four-sided spiral, where each face has the same width w:



Every slice still has an area of w^2, which means this figure has a volume of hw^2.

Exercise 15 Volume of a building

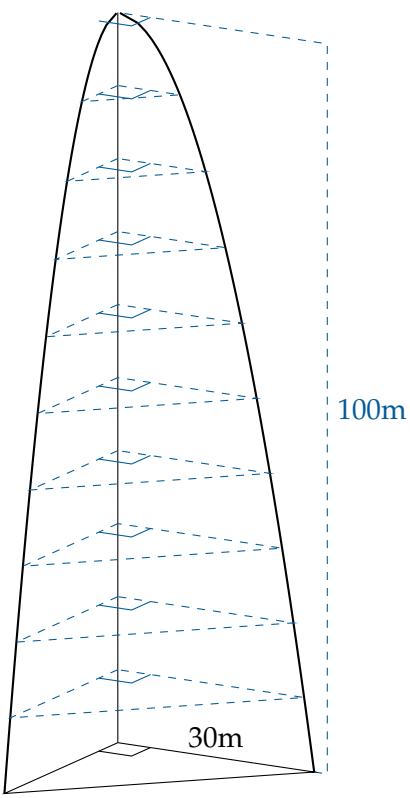
Working Space

An architect is designing a hotel with a right triangular base; the base is 30 meters on each leg. The building gets narrower as you get closer to the top, and finally shrinks to a point. The spine of the building is where the right angle is. That spine is straight and perpendicular to the ground.

Each floor has a right isosceles triangle as its floor plan. The length of each leg is given by this formula:

$$w = 30\sqrt{1 - \frac{z}{100}}$$

So, the width of the building is 30 meters at height $z = 0$. At 100 meters, the building comes to a point. It will look like this:



What is the volume of the building in cubic meters?

CHAPTER 4

Conic Sections

In mathematics, conic sections (or simply conics) are curves obtained as the intersection of the surface of a cone with a plane. The three types of conic section are the hyperbola, the parabola, and the ellipse. The circle is a special case of the ellipse, though historically it was sometimes called a fourth type. All of the equations below can be graphed on programs like Desmos.

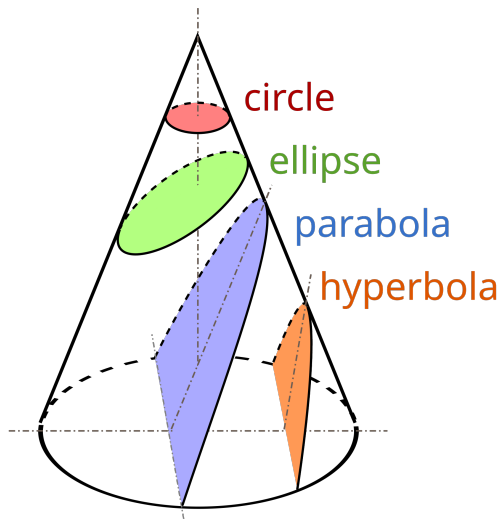


Figure 4.1: Visualization of conic sections.

Source: Wikimedia Commons, Public Domain: https://upload.wikimedia.org/wikipedia/commons/thumb/1/11/Conic_Sections.svg/1920px-Conic_Sections.svg.png

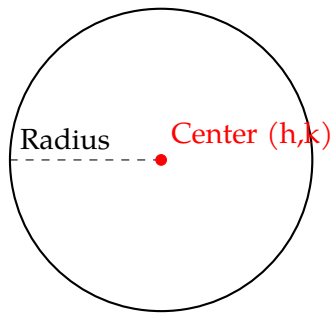
4.1 Definitions

Each type of conic sections can be defined as follows:

4.1.1 Circle

A circle is the set of all points in a plane that are at a given distance (the radius) from a given point (the center). The standard equation for a circle with center (h, k) and radius r is:

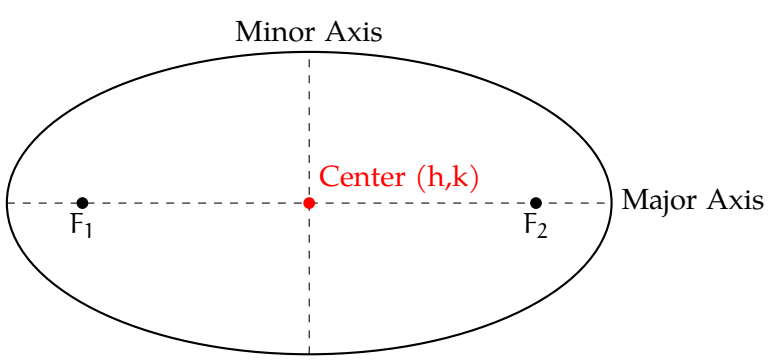
$$(x - h)^2 + (y - k)^2 = r^2 \tag{4.1}$$



4.1.2 Ellipse

An ellipse is the set of all points such that the sum of the distances from two fixed points (the foci) is constant. The standard equation for an ellipse centered at the origin with semi-major axis a and semi-minor axis b is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \tag{4.2}$$



4.1.3 Hyperbola

A hyperbola is the set of all points such that the absolute difference of the distances from two fixed points (the foci) is constant. A hyperbola is formed from slicing a *double-cone* — two cones placed tip-to-tip — parallel to or angled off of the central axes. The standard equation for a hyperbola centered at the origin is:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{4.3}$$

or

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1 \tag{4.4}$$

depending on the orientation of the hyperbola.

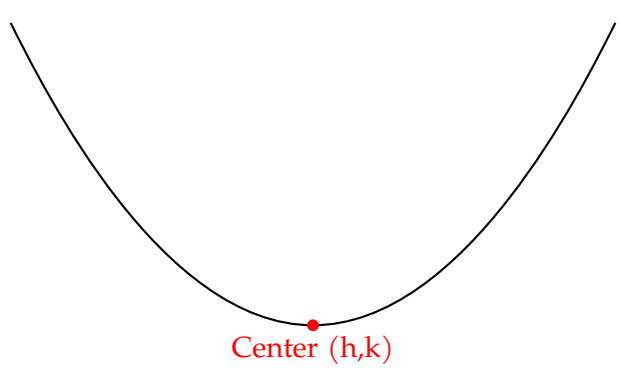
4.1.4 Parabola

A parabola is the set of all points that are equidistant from a fixed point (the focus) and a fixed line (the directrix). The standard equation for a parabola that opens upwards or downwards is:

$$y = a(x - h)^2 + k \tag{4.5}$$

and that opens leftwards or rightwards is:

$$x = a(y - k)^2 + h \tag{4.6}$$



where (h, k) is the vertex of the parabola, and a is a scalar.

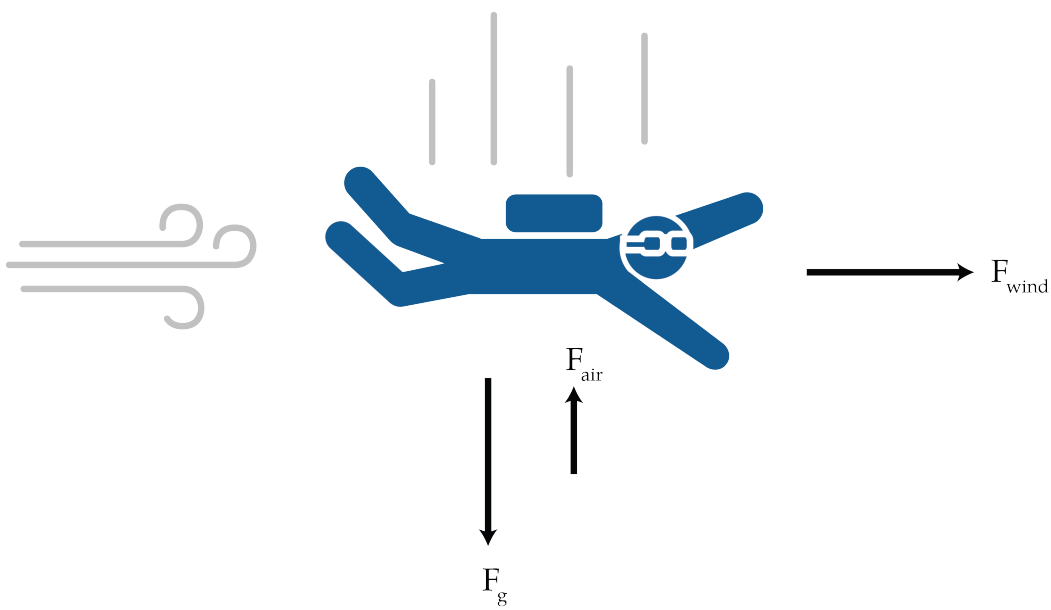
Note that only the parabola out of these four is a function, as it passes vertical line test. The other three cannot be expressed as functions, only equations.

Chapter 5

Vectors

We have talked a some about forces, but in the calculations that we have done, we have only talked about the magnitude of a force. It is equally important to talk about its direction. To do the math on things with a magnitude and a direction (like forces), we need vectors.

For example, if you jump out of a plane (hopefully with a parachute), several forces with different magnitudes and directions will be acting upon you. Gravity will push you straight down. That force will be proportional to your weight. If there were a wind from the west, it would push you toward the east. That force will be proportional to the square of the speed of the wind and approximately proportional to your size. Once you are falling, there will be resistance from the air that you are pushing through — that force will point in the opposite direction from the direction you are moving and will be proportional to the square of your speed.



To figure out the net force (which will tell us how we will accelerate), we will need to add these forces together. To do this, we need to learn to do math with vectors.

5.1 Adding Vectors

A vector is typically represented as a list of numbers, with each number representing a particular dimension. For example, if you are creating a 3-dimensional vector representing a force, it will have three numbers representing the amount of force in each of the three axes. For example, if a force of one newton is in the direction of the x-axis, you might represent the vector as $v = [1, 0, 0]$. Another vector might be $u = [0.5, 0.9, 0.7]$. You can see examples of 2-dimensional and 3-dimensional vectors in figures 5.1 and 5.2.

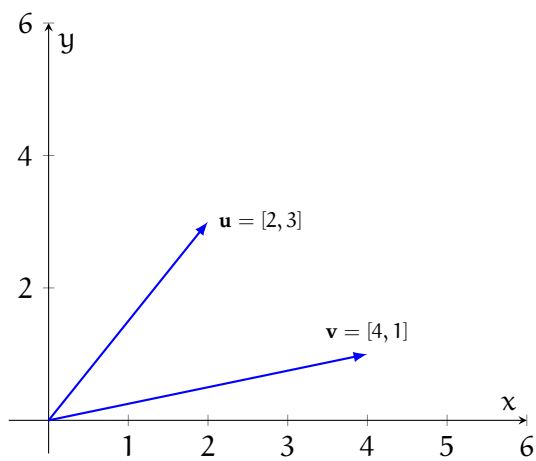


Figure 5.1: 2-dimensional vectors, **u** and **v**

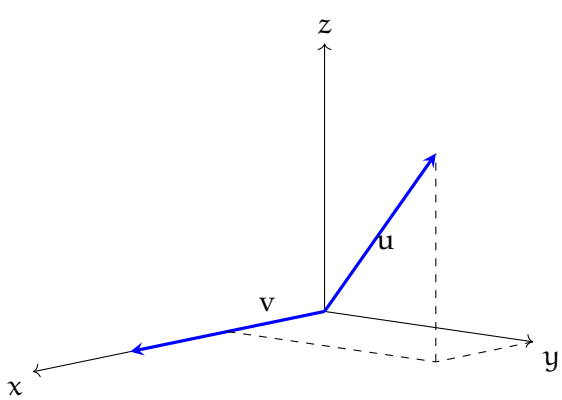


Figure 5.2: 3-dimensional vectors, **u** and **v**

Thinking visually, when we add to vectors, we put the starting point second vector at the ending point of the first vector. This is illustrated for 2-dimensional vectors in figure 5.3 and for 3-dimensional vectors in figure 5.4.

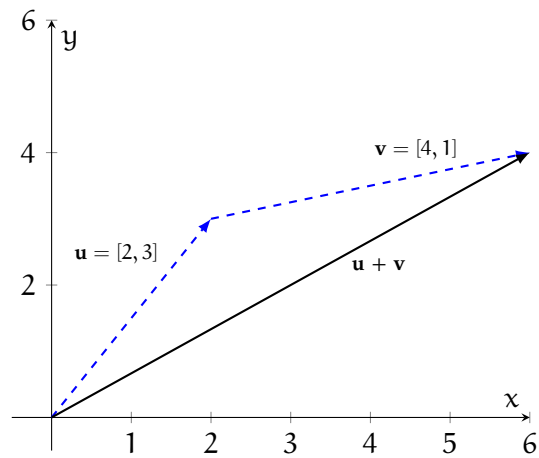


Figure 5.3: A visual representation of adding 2-dimensional vectors, **u** and **v**

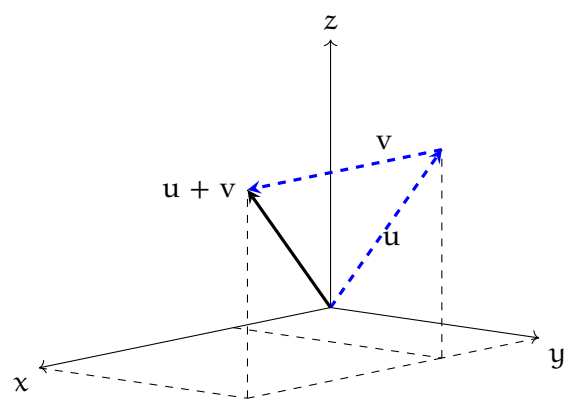


Figure 5.4: A visual representation of adding 3-dimensional vectors, **u** and **v**

If you know the vectors, you will just add them element-wise:

$$u + v = [0.5, 0.9, 0.7] + [1.0, 0.0, 0.0] = [1.5, 0.9.0.7]$$

These vectors have 3 components, so we say they are *3-dimensional*. Vectors can have any number of components. For example, the vector $[-12.2, 3, \pi, 10000]$ is 4-dimensional.

You can only add two vectors if they have the same dimension.

$$[12, -4] + [-1, 5] = [11, 1]$$

Addition is commutative; if you have two vectors a and b, then $a + b$ is the same as $b + a$.

Addition is also associative: If you have three vectors a, b, and c, it doesn't matter which order you add them in. That is, $a + (b + c) = (a + b) + c$.

A 1-dimensional vector is just a number. We say it is a *scalar*, not a vector.

Exercise 16 Adding vectors

Add the following vectors:

- $[1, 2, 3] + [4, 5, 6]$
- $[-1, -2, -3, -4] + [4, 5, 6, 7]$
- $[\pi, 0, 0] + [0, \pi, 0] + [0, 0, \pi]$

Working Space

Answer on Page 76

Exercise 17 Adding Forces

You are adrift in space, near two different stars. The gravity of one star is pulling you towards it with a force of $[4.2, 5.6, 9.0]$ newtons. The gravity of the other star is pulling you towards it with a force of $[-100.2, 30.2, -9.0]$ newtons. What is the net force?

Working Space

Answer on Page 76

5.2 Multiplying a vector with a scalar

It is not uncommon to multiply a vector by a scalar. For example, a rocket engine might have a force vector v. If you fire 9 engines in the exact same direction, the resulting force vector would be $9v$.

Visually, when we multiply a vector u by a scalar a, we get a new vector that goes in the

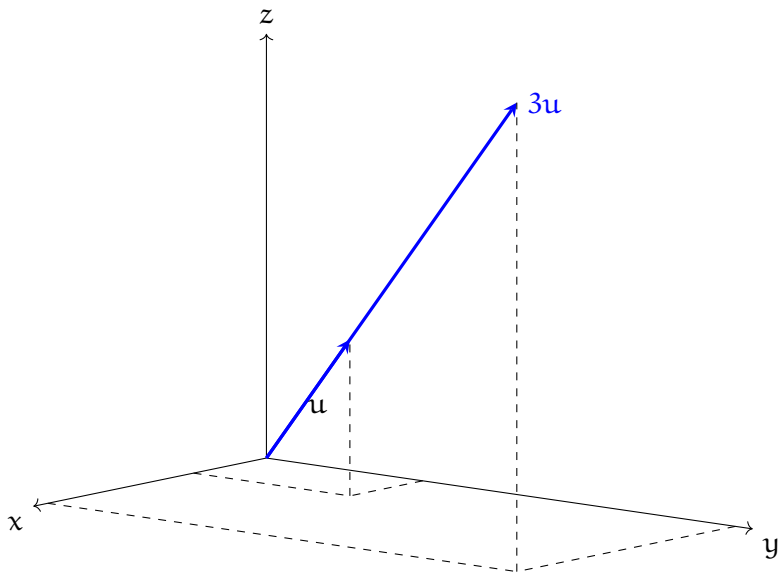


Figure 5.5: To multiply vectors, the vector gets stretched in the same direction a amount

same direction as u but has a magnitude a times as long as u. A visual is presented in figure 5.5.

When you multiply a vector by a scalar, you simply multiply each of the components by the scalar:

$$3 \times [0.5, 0.9, 0.7] = [1.5, 2.7, 3.6]$$

Exercise 18 Multiplying a vector and a scalar

Simplify the following expressions:

Working Space

- $2 \times [1, 2, 3]$
- $[-1, -2, -3, -4] \times -2$
- $\pi[\pi, 2\pi, 3\pi]$

Answer on Page 76

Note that when you multiply a vector times a negative number, the new vector points in

the opposite direction (see figure 5.6).

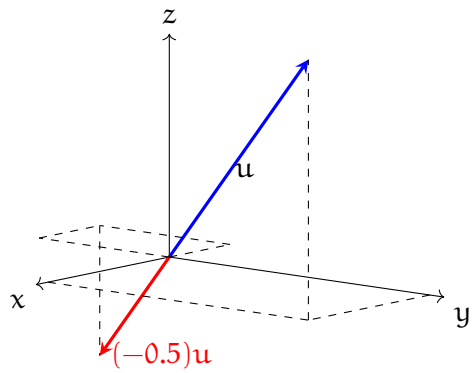


Figure 5.6: Multiplying a vector by a negative number reverses the direction of the vector.

5.3 Vector Subtraction

As you might guess, when you subtract one vector from another, you just do element-wise subtraction:

$$[4, 2, 0] - [3, -2, 9] = [1, 4, -9]$$

So, $u - v = u + (-1v)$.

Visually, you reverse the one that is being subtracted (see figure 5.7):

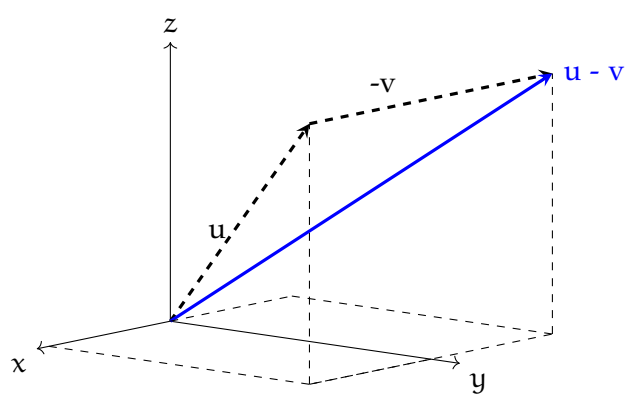


Figure 5.7: To subtract a vector, you reverse it, then add the reversed vector.

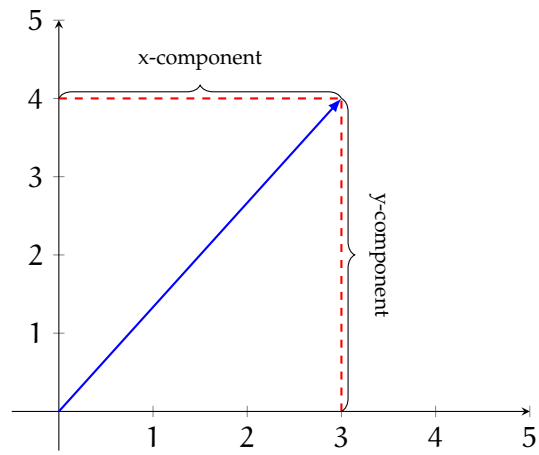


Figure 5.8: The magnitude of a vector can be thought of as the length of a hypotenuse of a right triangle.

5.4 Magnitude of a Vector

The *magnitude* of a vector is just its length. We write the magnitude of a vector v as $|v|$.

We compute the magnitude using the pythagorean theorem. If $v = [3, 4, 5]$, then

$$|v| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50} \approx 7.07$$

(You might notice that the notation for the magnitude is exactly like the notation for absolute value. If you think of a scalar as a 1-dimensional vector, the absolute value and the magnitude are the same. For example, the absolute value of -5 is 5. If you take the magnitude of the one-dimensional vector $[-5]$,you get $\sqrt{25} = 5$.)

Where does this equation come from? Consider a 2-dimensional vector, $\mathbf{v} = [3, 4]$. This means the the vector represents 3 units in the x-direction, and 4 units in the y-direction. We can then visualize a right triangle, with the vector being the hypotenuse and the legs being the x- and y-components of the vector (see figure 5.8). As you recall, the length of the hypotenuse of a right triangle is the square root of the sum of the squares of the legs. That is:

$$c = \sqrt{a^2 + b^2}$$

Where c is the length of the hypotenuse and a and b are the lengths of the legs. See Figure **??**.

We won't prove it here, but this method holds for higher-dimension vectors as well.

Magnitude of Vectors

For an n-dimensional vector, $\mathbf{v} = [x_1, x_2, x_3, \cdots, x_n]$, the magnitude of the vector is given by:

$$|\mathbf{v}| = \sqrt{x_1^2 + x_2^2 + x_3^2 + \cdots + x_n^2}$$

Notice that if you scale up a vector, its magnitude scales by the same amount. For example:

$$|7[3, 4, 5]| = 7\sqrt{50} \approx 7 \times 7.07$$

Here is why that is true. Suppose we have a vector, $\mathbf{u} = [a, b, c]$. Then the magnitude of $\mathbf{u}$ is given by:

$$|\mathbf{u}| = \sqrt{a^2 + b^2 + c^2}$$

If we scale $\mathbf{u}$ to create $\mathbf{v}$ such that $\mathbf{v} = k\mathbf{u} = [ka, kb, kc]$, where k is some constant. Then the magnitude of $\mathbf{v}$ is given by:

$$|\mathbf{v}| = \sqrt{(ka)^2 + (kb)^2 + (kc)^2}$$

We can expand and simplify this equation:

$$|\mathbf{v}| = \sqrt{k^2a^2 + k^2b^2 + k^2c^2}$$

$$|\mathbf{v}| = \sqrt{k^2\left(a^2 + b^2 + c^2\right)}$$

$$|\mathbf{v}| = \left(\sqrt{k^2}\right)\sqrt{a^2 + b^2 + c^2}$$

$$|\mathbf{v}| = |k|\sqrt{a^2 + b^2 + c^2} = |k|\,|\mathbf{u}|$$

So, if you scale a vector, the magnitude of the resulting vector is the absolute value of the scale factor times the magnitude of the original vector.

The rule then is: If you have any vector v and any scalar k:

$$|kv| = |k||v|$$

5.4.1 Unit Vectors

A *unit vector* is a vector whose magnitude is 1. For any non-zero vector $\vec{\mathbf{v}}$, the unit vector $\vec{\mathbf{u}}$ pointing in the same direction is

$$\vec{\mathbf{u}} = \frac{\mathbf{v}}{|\mathbf{v}|}.$$

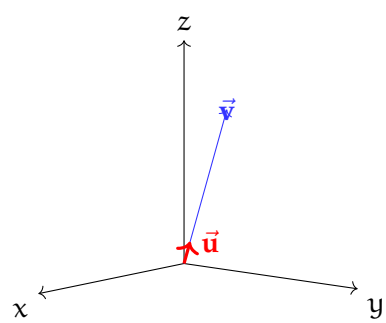
For example, if $\mathbf{v} = [3, 4, 5]$ then

$$|\mathbf{v}| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50},$$

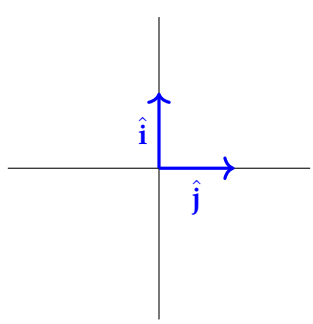
so

$$\vec{\mathbf{u}} = \frac{1}{\sqrt{50}}[3, 4, 5] = \left[\tfrac{3}{\sqrt{50}}, \tfrac{4}{\sqrt{50}}, \tfrac{5}{\sqrt{50}}\right].$$

$$\boxed{\widehat{\mathbf{u}} = \frac{\mathbf{v}}{|\mathbf{v}|}}, \qquad |\vec{\mathbf{u}}| = 1.$$

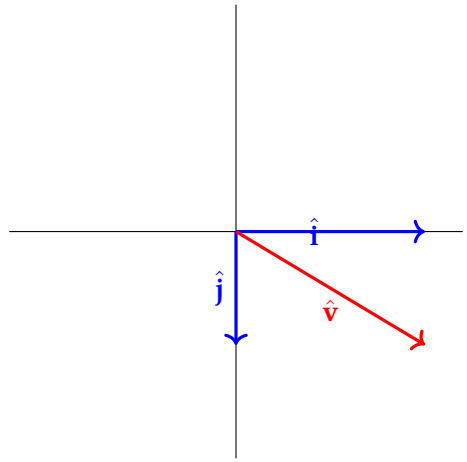


Unit vectors are often represented by placing a hat over the vector. In standard vector calculus, the variables $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{i}}$ for the x, y, and z variables respectively.



You may see vectors provided to you in the form $\mathbf{v} = 5\hat{\mathbf{i}} - 3\hat{\mathbf{j}}$. You may say this as '5 times the unit vector in the x-direction plus the -3 times the unit vector in the y-direction'. The

same may apply for 3D vectors as well.



Exercise 19 Magnitude of a Vector

Find the magnitude of the following vectors:

- $[1, 1, 1]$
- $[-5, -5, -5]$ (that is the same as $-5 \times [1, 1, 1]$)
- $[3, 4, -4] + [-2, -3, 5]$

Working Space

Answer on Page 77

5.5 Vectors in Python

NumPy is a library that allows you to work with vectors in Python. You might need to install it on your computer. This is done with `pip`. `pip3` installs things specifically for Python 3.

```
pip3 install NumPy
```

We can think of a vector as a list of numbers. There are also grids of numbers known as *matrices*. NumPy deals with both in the same way, so it refers to both of them as arrays.

The study of vectors and matrices is known as *Linear Algebra*. Some of the functions we need are in a sublibrary of NumPy called `linalg`.

As a convention, everyone who uses NumPy, imports it as *np*.

Create a file called `first_vectors.py`:

```
import NumPy as np

# Create two vectors
v = np.array([2,3,4])
u = np.array([-1,-2,3])
print(f"u = {u}, v = {v}")

# Add them
w = v + u
print(f"u + v = {w}")

# Multiply by a scalar
w = v * 3
print(f"v * 3 = {w}")

# Get the magnitude
# Get the magnitude
mv = np.linalg.norm(v)
mu = np.linalg.norm(u)
print(f"|v| = {mv}, |u| = {mu}")
```

When you run it, you should see:

```
> python3 first_vectors.py
u = [-1 -2  3], v = [2 3 4]
u + v = [1 1 7]
v * 3 = [ 6  9 12]
|v| = 5.385164807134504, |u| = 3.7416573867739413
```

5.5.1 Formatting Floats

The numbers 5.385164807134504 and 3.7416573867739413 are pretty long. You probably want them rounded off after a couple of decimal places.

Numbers with decimal places are called *floats*. In the placeholder for your float, you can

specify how you want it formatted, including the number of decimal places.

Change the last line to look like this:

```
print(f"|v| = {mv:.2f}, |u| = {mu:.2f}")
```

When you run the code, it will be neatly rounded off to two decimal places:

```
|v| = 5.39, |u| = 3.74
```

Chapter 6

Charge

If you rub a balloon against your hair, then place it next to a wall, it will stick. Why does this happen? This is exactly what will be explored in this chapter on **charge** and **electrostatics**. This is because the balloon has stripped some electrons from your hair, and now the ballon has slightly more electrons than protons. We say that it has gotten an *net electrical charge*, or an imbalance between protons and electrons. While any **friction** or contact between two objects can cause a difference in electric charge, *net charge cannot be created or destroyed*. In this case, the balloon has a negative electrical charge. Objects with slightly more protons than electrons have a positive charge. Consequently, your hair now has a positive charge. Note that the differences are equal but opposite.

An object with a negative charge and an object with a positive charge will be *attracted* to each other. Two objects with the same charge will be *repelled* by each other. This charge originates from a subatomic level, when protons (positively charged) and electrons (negatively charged) are present in an atom at *unequal amounts*.

This charge, referred to as *elementary charge* is represented by the base unit e (not to be confused with the e^x function in math disciplines). This value, known as the elementary charge (e), represents the smallest indivisible unit of electric charge. Much like counting physical coins, you cannot have a "fractional" coin. Similarly, an object's net charge must be an integer multiple of e. In physics, we describe this property by saying that charge is quantized ($Q = ne$ where n is an integer).

In this way, we call elementary charge *quantized*. This was shown back in Because it is quantized, we represent the charge of a particle as q. The SI (International System of Units) measures charge in *coulombs*. The charge of a single proton is about 1.6×10^{-19} coulombs.

Two objects with charges are bound to either attract (pull) or repel (push) on each other. This is a type of force referred to as *electric force*.

Coulomb's Law

If two objects with charge q_1 and q_2 (in coulombs) are r meters from each other, the force of attraction or repulsion (electric force) is given by

$$F = k\frac{q_1 q_2}{r^2} \tag{6.1}$$

where F is in newtons and K is Coulomb's constant: about 8.988×10^9. You may use $9 \times 10^9\,\mathrm{N \cdot m^2/C}$ in your calculations.

This k value comes from $\frac{1}{4\pi\epsilon_0}$, a *relationship with the permittivity of free space*, ϵ_0.[1] You may see this as F_e, F_q, or F_c, but they all refer to the electric force and the standard is F_e.

Note how similar this looks to Newton's Law of Universal Gravitation:

$$F_g = G\frac{m_1 m_2}{r^2}$$

Both laws follow an inverse-square relationship, meaning the force decreases with the square of the distance. However, gravity is always attractive, while electric force can be either attractive or repulsive depending on the signs of the charges (q_1 and q_2).

If this force is negative, it represents the idea that the two given particles are attracted to each other, as this would mean one of the particles must have a negative charge.

If this force is positive, it would mean there is a push or repelling force on each of the particles.

What if this force is zero? This would only result if one or both particles have zero charge. This would mean that they do not have an electric force, $F_E = 0$.

Exercise 20 Coulomb's Law

Two balloons are charged with an identical quantity and type of charge: -5×10^{-9} coulombs. They are held apart at a separation distance of 12 cm. Determine the magnitude of the electrical force of repulsion between them.

Working Space

Answer on Page 77

[1] A note on the permittivity of free space: ϵ_0 (or the electric constant) represents the capability of a vacuum to permit electric field lines. Its exact value in SI units is $\epsilon_0 \approx 8.854 \times 10^{-12}\ \mathrm{C^2/(N \cdot m^2)}$. This tells us that the speed of light isn't just a number but rather it is strictly determined by how quickly the universe can allow electric and magnetic fields to change. Using k allows for quick math, especially when fields become spherical and the π factor is eliminated easily.

At this point, you might ask "If the wall has zero charge, why is the balloon attracted to it?" The answer: the electrons in the wall move away from the balloon, polarizing the atoms. The negative charge on the balloon pushes electrons away from itself, so the surface of the wall gets a mild positive charge. The negative charge on the balloon is attracted more to the positive charge on the surface of the wall than the negative charge on the inside, thus the balloon sticks. This is due to the idea of *conductors* and *insulators*.

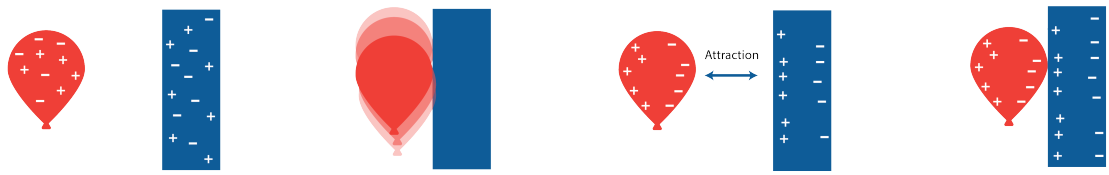


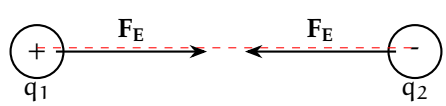
Figure 6.1: The negative balloon repels the wall's electrons, leaving a net positive surface. The balloon is then attracted to this induced positive charge.

Recall that a **conductor** is a type of material through which valence electrons (electrons in the outer shell) easily flows, and an **insulator** is a type of material through which charge does not easily flow, if at all. Metals, such as copper or zinc, are the best conductors because the electrons within the atoms are not tied to a nucleus, but rather form a sparse collection or sea in which the charge can flow through. Insulators like plastic, rubber, or glass are prevent charge flowing through as the electrons are tightly bound to their parent atoms or molecules.

6.0.1 Point Charge

The wall and the balloon all contain a bunch of tiny particles which we can refer to as **point charges**. A point charge is an electric charge that is treated as a concentrated single point in space, with no physical size or volume. Similar to point masses, we use point charges to simulate effects of charge on other objects or points.

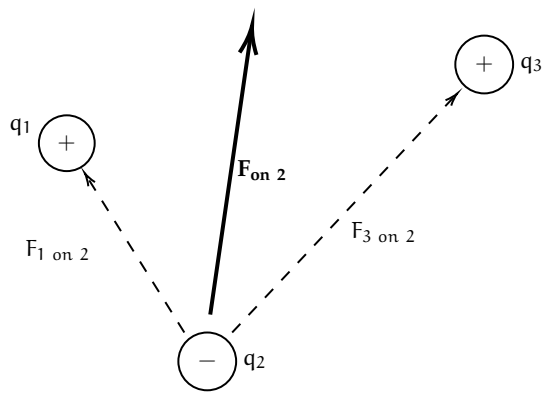
This charge acts parallel to the secant line that touches both point charges only once.



6.0.2 Superposition

What if there are more just one charges acting on a given point charge? Coulomb's law gives the force between two point charges, but in a system with three or more charges we apply the **principle of superposition**[2].

The principle of superposition states that the total electric force on a charge is the vector sum of the individual forces exerted on it by all other charges.[3]



Exercise 21 Finding net charge on a point charges

Working Space

This question is taken from the AP Physics C 19th edition practice workbook.

Consider four equal, positive point charges that are situated at the vertices of a square. Find the net electric force on a negative point charge placed at the square's center.

Hint: draw a diagram to help you

Answer on Page 77

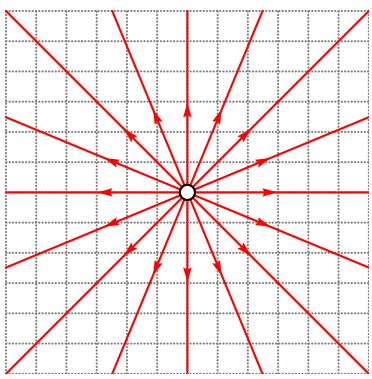
[2]Superposition is really a linear algebra topic, and not something that this textbook will cover, but it has so many real life applications and we recommend you do your own research! See the wiki page for more: `https://en.wikipedia.org/wiki/Superposition_principle`

[3]This is similar how we calculate tensions when there are multiple tensions (or just forces in general) on an object. The net force is the sum of all the forces after vector addition is applied.

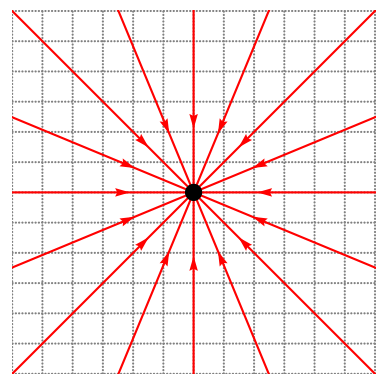
6.1 Electric Field

Recall that, when we talked about Gravitational Fields, we defined them as fields of infinitesimal points that each have a magnitude of force associated with them. This field revolves around the entire 2D circle or 3D sphere (generally point masses, charges, fields, etc are spherical).

Similar to how the gravitational field extends out from the center of a given point mass, an *electric field* of a point charge has (imaginary) vectors pointing out of it. However, the direction of the vectors[4] (and electric force) depends on the the charge of the point charge. A negatively charged point mass has force lines pointing *towards* it, and a positively charged point mass has force lines pointing *away* from it (6.2a). In short, they always originate from positive charges and approach negative charges.



(a) A positively (open circle) charged electric field with corresponding electric field lines.



(b) A negatively (filled circle) charged electric field with corresponding electric field lines.

Figure 6.2: Two electric fields with different charges.

What are the *units* of an electric field? Well, by this equation, $\vec{\mathbf{F}}_e$ is a force which we know is in *Newtons* (N), and the test particle q is in Coulombs (C). The resulting units are $\mathrm{N\,C^{-1}}$, or Newton's per Coulomb. Note that this is equivalent to $1\,\mathrm{V\,m^{-1}}$, or one Volt per meter[5]. This is how electric field forces are converted to volts within batteries or circuits!

It is important to note that we can simplify this this equation by substituting Coulomb's law:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_e}{q_t} = \frac{\frac{kq_1q_t}{r^2}}{q_t} = \frac{kQ}{r^2} \tag{6.2}$$

where Q is the electric field's source charge.

[4]You may hear the terms electric field lines and electric field vectors used interchangibly

[5]Equivalently, one $\mathrm{kV\,cm^{-1}}$, or kilovolt per centimeter is equal to $10^{-5} \times 1\mathrm{N\,C^{-1}}$. To reverse the operation, multiply the kilovolts per centimeter by 10^5 to get the resulting Newtons per Coulomb

For positive charges Q, the force vectors get weaker the farther from the charge (visually, the force lines get shorter). In fact, it decreases by a factor of $\frac{1}{r^2}$. Think to yourself where this factor comes from. (Hint: refer back to Coulomb's law (6.1))

Electric Field Force from a singular charge to a second charge

A charge q_1 creates an electric field at a distance r from it with strength:

$$E = \frac{kq_1}{r^2} \tag{6.3}$$

Introducing a new charge q_2, the force on it due to the electric field is:

$$F = q_2 \cdot \frac{kq_1}{r^2} \iff F = q_2 E \tag{6.4}$$

where E is the electric field strength at the location of q_2. This is often referred to as the charge within a known field.

Exercise 22 Point Charges and Electric Fields

This problem is adapted from an Example IB Exam. Two isolated point charges, X of charge +Q and Y of charge +2Q, are separrated by distance 3d. Point P is distance d from X and distance 2d from Y.

What is the net electric field strength at P? Are the forces from each charge equal?

1. 0, equal forces
2. $\frac{2kQ}{d^2}$, equal forces
3. $\frac{kQ}{2d^2}$, equal forces
4. $\frac{2kQ}{1.5d^2}$, equal forces

Working Space

Answer on Page 77

The density of the electric force lines at a given point determines the strength of the force at the point. Locations where field lines are more dense are much stronger than areas where the lines are sparse. This goes back to the idea of the force strength of an electric field.

If you recall the properties of conductors: *there can be no electrostatic field within the body of a conductor*. If an electric field were to exist inside the conductor, it would exert a force $F = qE$ on these free charges, causing them to accelerate. This internal motion continues until the charges redistribute themselves along the conductor's outer surface, creating an induced internal field that perfectly opposes and cancels the original field.

Electric field lines always

- Start at positive charges and terminate at negative ones. If it is a single charge, this is indicated by the direction of arrows off to infinity.
- The number of lines is proportional to the magnitude of charge.
- The line approaches a conductive surface perpendicularly.
- No two field lines ever cross in a charge-free region.

Keep these in mind for any sketches you may do.

6.2 Dipoles

An **electric dipole** is a set of two *point charges* that equal and opposite charges that are infinitesimally close together. Two point charges, one with charge $+q$ and the other one with charge $-q$ separated by a distance d, create an electric dipole.

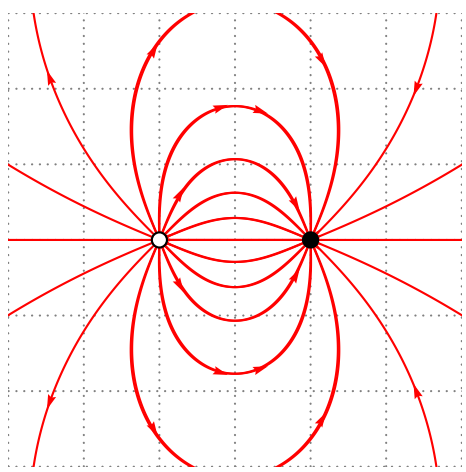


Figure 6.3: A basic dipole diagram.

Charge flows consistently from positive to negative, and has magnitude $\vec{\mathbf{p}} = q\vec{\mathbf{d}}$, where $\vec{\mathbf{p}}$ is the dipole moment with units $C \cdot m$, or Coulomb-meters.

6.3 Electric Flux and Gauss's law

Electric Flux is the number of field lines that pass through a given surface. Represented by Φ, it is the measure of how much of an electric field itself passes through an area.

The electric flux depends on 3 factors:

- Electric Field Strength $(\mathbf{E})$
- Surface Area being measured, $(\mathbf{A})$. $\mathbf{A}$ must be perpendicular to the field lines.
- Angle at which the area is relative to the field lines.

If the electric field lines are perpendicular to the area $\mathbf{A}$, then the electric flux is

$$\Phi = \sum \mathbf{E} \cdot \mathbf{A} \tag{6.5}$$

Think of this as a paper directly perpendicular to a fan; the maximum amount of the paper is blown by the fan.

If the paper is at an angle to the fan, less flow from the fan affects the paper. If you tilt it until it's parallel to the wind, zero wind affects the paper. Similarly, the electric flux on a surface at a titled angle decreases how much flux is felt:

$$\Phi = \sum \mathbf{E} \cdot \mathbf{A} \cos(\theta) \tag{6.6}$$

In integral form (which we have not covered yet, so do not worry about using it), this is:

$$\Phi = \oint \mathbf{E} \cdot d\mathbf{A} \cos(\theta) \tag{6.7}$$

where the integral is on a closed area.

If you recall our discussion on resistance, flux and resistance are very similar measurements. In a wire, a larger cross-sectional area (A) is like a highway for electrons. It allows more electrons to pass through easily, which decreases the resistance. If you maximize flux by having a huge area, you are essentially creating a path with very low resistance for a current.

6.3.1 Gauss's law

So far, we have imagined the electric field is only 2-dimensional plane (a circle). What if we consider them as a 3-dimensional boundary, similar to gravitation? This introduces the idea of *Gauss's Law for Electric Fields*:

Gauss's Law

Gauss's law states that the strength of an electric field for 3-Dimensional object can be calculated by:

$$\Phi = \oiint \vec{\mathbf{E}}\, d\vec{\mathbf{A}} \cos(\theta) = \frac{Q_{enc}}{\epsilon_0} \tag{6.8}$$

This is a formula for saying that every infinitessimal bit of a 2-Dimensional object is given a height and an electric strength, and this is equal to the enclosed area of the charge over the permittivity of free space constant.

For specifically spherically 3-Dimensional objects, inputting the formula for a sphere into A gives us:

$$E = \frac{q}{4\pi r^2 \epsilon_0} \tag{6.9}$$

where q is the source charge, r is the distance from that source charge, and $4\pi r^2$ is the surface area of the electric field. What this means is that electric field at all points r from the source charge are equal in magnitude. Pretty cool!

6.4 Continuous Charge Distributions

In the previous sections, we treated charges as discrete "dots" in space. However, in applications of charges such as a long charged wire, a flat metal plate, or a solid plastic sphere, charge is spread out over a macroscopic object. When we have a collection of so many point charges that they appear to form a smooth, "fluid-like" surface or volume, we call it a **continuous charge distribution**. This means that the charges, while not truly continuous, can be considered continuous as a long single stream.

To analyze these, we stop summing individual particles ($\sum$) and instead use calculus to integrate ($\int$) over the entire object. While we have not covered calculus yet, keep in mind that this is just a way of calculating the area or volume of a continuous solid.

6.4.1 Charge Densities

The first step in any continuous distribution problem is defining how the charge is spread. We use three Greek letters to represent the "concentration" of charge based on the object's

geometry:

Type	Symbol	Definition	SI Units
Linear	λ (lambda)	Charge per unit length	C/m
Surface	σ (sigma)	Charge per unit area	C/m^2
Volume	ρ (rho)	Charge per unit volume	C/m^3

Table 6.1: The three types of charge density used in electrostatics.

If the charge is distributed **uniformly**, these values stay constant. For example, for a rod of length L and total charge Q, the linear density is simply $\lambda = Q/L$. However, if these the charge is not constant per the unit of length, we must use calculus to find the total charge density. To find the total electric field $\vec{\mathbf{E}}$ at a point P near a charged object, we imagine breaking the object into many infinitesimal "pieces" of charge, dq. Each chunk acts like a tiny point charge and contributes a tiny bit of electric field, $d\vec{\mathbf{E}}$.

Recall our point charge formula:

$$d\vec{\mathbf{E}} = \frac{1}{4\pi\epsilon_0}\frac{dq}{r^2} \tag{6.10}$$

To find the total field, we integrate over the entire distribution:

$$\vec{\mathbf{E}} = \int d\vec{\mathbf{E}} = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r^2} \tag{6.11}$$

The key to solving these integrals is replacing dq with the appropriate density:

- Lines: $dq = \lambda\, dl$
- Surfaces: $dq = \sigma\, dA$
- Volumes: $dq = \rho\, dV$

For now, you will not be quizzed on these integrals but know of their existence and that charge can be treated as a virtual stream along an object. This is why any part of the balloon from our first example will stick to any part of the wall.

Exercise 23 Belt being charged

Working Space

This question is from an IB Electromagnetics problem. A continuous conveyor belt of width 0.60 m travels at a constant speed of $2.5\,\mathrm{ms}^{-1}$. The belt has a uniform distribution of charge of $5.0\mathrm{mCm}^{-2}$ on the surface. As the belt passes a point A, all the charge is transferred into the wire with width along the horizontal of the belt.

What is the charge on the wire? *Hint: recall that current is* $\frac{\Delta Q}{\Delta t}$.

Answer on Page 78

6.5 Electric Potential

Let's talk about electric potential and electric potential energy. Recall that objects can have *kinetic* and *potential* energy, that potential energy is gradually transported into kinetic energy.

Electric Potential describes the amount of work, W_E, done by an electric force. The change in **electric potential energy** is defined as the work done on that force:

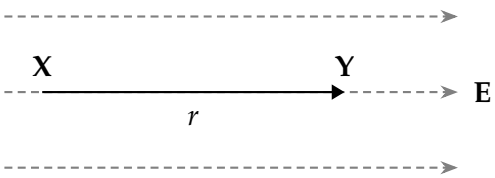
$$\Delta U_E = -W_E \tag{6.13}$$

Notice how this is similar to an object of mass m undergoing work done by a gravitational field $\Delta U_G = -W_G$.

Exercise 24 Change in Electric Potential Energy

Working Space

A positive charge $+q$ moves a distance r from position X to position Y in a **uniform** electric field E.



What is its change in electric potential energy? Assume moving in direction r moves in the direction of E.

Answer on Page 78

Electric Potential (V) is the amount of work needed to move a test charge q_t from a reference point to a specific point in a static electric field. It describes the potential for a charge to have energy at that location, regardless of whether a charge is actually there. Electric Potential Energy (U_E) per unit charge where ($V = U_E/q$).

Just as moving a mass up in height against gravity increases its gravitational potential, moving a positive charge "up" against an electric field increases its electric potential. In both cases, the field *tries* to push the object toward a state of lower potential energy. A positive charge naturally wants to move with the electric field toward lower potential; in this case, its electric potential energy is converted into kinetic energy. Conversely, to move a positive charge against an electric field, work must be done on the charge, causing its electric potential to increase.

Moving from two points not on the same electric field line introduces an angle component θ:

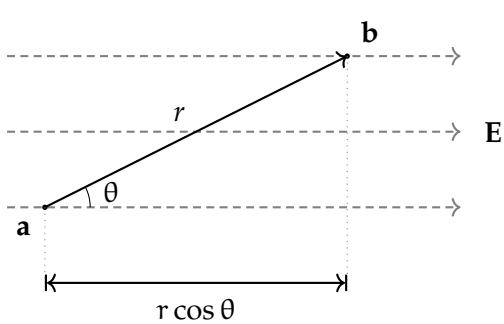


Figure 6.4: A gain of potential energy across an angle.

The electric potential difference between two points A and B is defined by the line

integral of the electric field:

$$\Delta V = V_B - V_A = -\int_A^B \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} \tag{6.14}$$

The negative sign indicates that the potential decreases in the direction of the electric field.

We can easily define two point charges q_1 and q_2 separated by distance r to have potential energy

$$U = k\frac{q_1 q_2}{r} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r} \tag{6.15}$$

Exercise 25 Accelerating an Electron

This is a sample problem from an IB Physics Exam. An electron is accelerated from rest by an electric potential difference V. The electron reaches a speed of 9.4×10^6 m/s. Using the charge of an electron as $e = 1.6 \times 10^{-19}$ C and the mass of an electron as $m_e = 9.11 \times 10^{-31}$ kg, find the potential difference V that accelerated the electron to this speed. You may use a calculator to simplify computations.

Answer on Page 79

6.5.1 Point Charges, Uniform Fields, and Spherical Shells

In a **uniform electric field** (such as the space between two large parallel plates), the field strength E is constant. The potential difference between two points separated by a distance d along the field lines is:

$$\Delta V = -Ed \tag{6.16}$$

For a **point charge** Q, we typically set the reference point ($V = 0$) at infinity. The potential at a distance r from the charge is:

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{r} \tag{6.17}$$

Unlike the electric field, which is a vector and follows an $1/r^2$ factor, electric potential is a **scalar** and follows an $1/r$ factor.

Surface Spherical Charge

For all spherical objects of radius R (surface spheres - assume hollow interior), it has potential inside and outside of the shell. For all points outside the shell, it is the same for any point charge Q, as seen in equation 6.17. For all points *inside* the shell, the potential is

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{R}$$

This means that moving a test charge q from the surface of the shell to the center of the shell, the potential does not change. The potential is constant within the shell.

When the sphere is *not hollow*, the interior charge changes to:

$$k = \frac{1}{4\pi\epsilon_0}\frac{Q}{2R}\left(3 - \frac{r^2}{R^2}\right)$$

6.5.2 Potential due to Configurations of Charge

Because potential is a scalar, finding the total potential from multiple charges is often simpler than finding the total electric field. We do not need to worry about components, instead, we simply sum the potentials:

$$V = \sum_i \frac{kq_i}{r_i} \quad \text{or for continuous distributions} \quad V = k\int \frac{dq}{r} \tag{6.18}$$

6.5.3 Conservation of Energy

When a charge q moves through a potential difference ΔV, the change in its electric potential energy is $\Delta U_E = q\Delta V$. If no other forces are acting on the charge, the total mechanical energy is conserved:

$$K_a + U_{E,a} = K_b + U_{E,b} \implies \frac{1}{2}mv_a^2 + qV_a = \frac{1}{2}mv_b^2 + qV_b \tag{6.19}$$

6.5.4 Summary

Particle	*wants* to move toward...	Potential Change (ΔV)	Potential Energy Change (U_E)
Positive ($+q$)	Lower Potential	Negative (Decreases)	Negative (Decreases)
Negative ($-q$)	Higher Potential	Positive (Increases)	Negative (Decreases)

Exercise 26 Electron between Field Plates

This is a sample problem from an IB Physics Exam. An electron that is travelling at a horizontal speed $v_x = 9.4 \times 10^6 \text{ms}^{-1}$ now enters a region of a *uniform electric field* between two parallel charged plates, separated by a distance of 4.0 cm. The electron is initially halfway between the plates and its initial velocity is parallel to the plates. The upper plate is positively charged and the lower plate is negatively charged. There is a potential difference between the plates of $V = 30\,\text{V}$.

Find the horizontal distance travelled before hitting a plate. Which plate does the electron hit?

Working Space

Answer on Page 79

6.6 Lightning

A cloud is a cluster of water droplets and ice particles. These droplets and ice particles are always moving up and down through the cloud. Recall that heat rises, so water is often trying to get higher in the cloud. In this process, electrons get stripped off and end up on the water droplets at the bottom of the cloud (water droplets collect at the bottom because they are denser). The air between the droplets is a pretty good insulator, which means the electrons are reluctant to jump anywhere. However, eventually, the charge gets so strong that even the insulating properties of the air is not enough to prevent the jump to earth, causing lightning. *Storms* are dense in clouds, so there is constant jumping of the electrons, which is why storms are prone to produing so much lightning. These electrons really want to jump to earth, and when given enough time and ability to bypass the air's

density. The electrons energy is visible as light and heat, and so we see a spark in the shape of a jagged line hitting a single point, the most conductive point on earth closest to the cloud. A great deal of lightning moves within a cloud or between clouds. However, sometimes it jumps to the earth. These bolts of lightning vary in the amount of electrons they carry, but the average is about 15 coulombs.

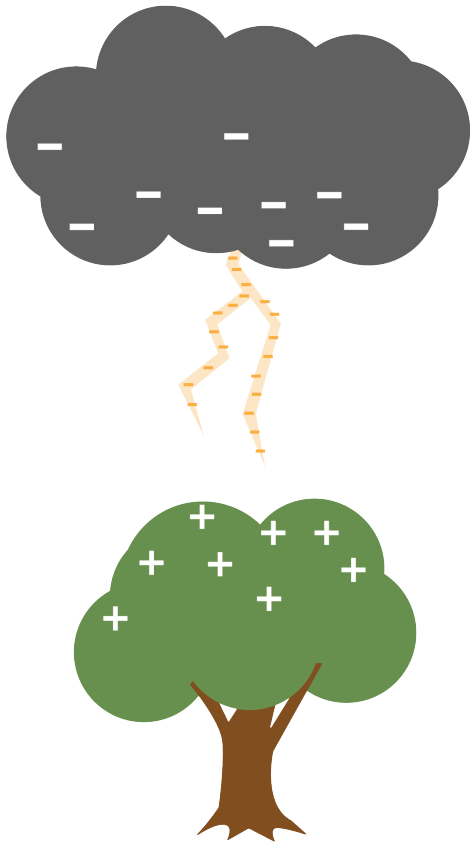


Figure 6.5: A difference of charge causes a spark of lightning to generate and transfer.

Thunder occurs because the electrons heat the air they pass through, causing the air to expand suddenly. The resulting shockwave is the sound we know as thunder.

6.7 But...

This idea that opposite charges attract creates some heavy questions that you do not yet have the tools to work with. So to these questions, the answer is basically "Don't ask that yet!"

However, you probably have these questions, so we will point you in the direction of the

answers.

The first is "In any atom bigger than hydrogen, there are multiple protons in the nucleus. Why don't the protons push each other out of the nucleus?"

We aren't ready to talk about it, but there is a force called *the strong nuclear force*, which pulls the protons and neutrons in the nucleus of the atom toward each other. At very, very small distances, it is strong enough to overpower the repulsive force due to the protons' charges.

Another question is "Why do the electrons whiz around in a cloud so far from the nucleus of the atom? Negatively charged electrons should cling to the protons in the center, right?"

We aren't ready to talk about this either, but quantum mechanics tells us that electrons like to live in a certain specific energy level. Hugging protons isn't one of those levels.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 7)

We start by writing out the limit:

$$\frac{d}{dx}\cos x = \lim_{h\to 0}\frac{\cos x + h - \cos x}{h}$$

Applying the sum formula for $\cos(x+h)$, we get:

$$= \lim_{h\to 0}\frac{\cos x\cos h - \sin x\sin h - \cos x}{h}$$

Rearranging to group the $\cos x$ and applying the Difference Rule:

$$= \lim_{h\to 0}\frac{\cos x\cos h - \cos x}{h} - \lim_{h\to 0}\frac{\sin x\sin h}{h}$$

Applying the Constant Multiple Rule:

$$= \cos x\lim_{h\to 0}\frac{\cos h - 1}{h} - \sin x\lim_{h\to 0}\frac{\sin h}{h}$$

Recalling that $\lim_{h\to 0}\frac{\sin h}{h} = 1$,

$$= \cos x\lim_{h\to a}\frac{\cos h - 1}{h} - \sin x\cdot 1$$

Recalling that $\lim_{h\to 0}\frac{\cos h - 1}{h} = 0$:

$$= \cos x\cdot 0 - \sin x = -\sin x$$

Therefore, $\frac{d}{dx}\cos x = -\sin x$

Answer to Exercise 2 (on page 7)

1. $\frac{\sec x(\tan x - 1)}{(1+\tan x)^2}$
2. $\sec t[\sec^2 t + \tan^2 t]$
3. $\frac{4 - \tan\theta + \theta\sec^2\theta}{(4-\tan\theta)^2}$
4. $2\sec t\tan t + \csc t\cot t$
5. $\frac{2}{(1+\cos\theta)^2}$
6. $\cos^2 x - \sin^2 x$

Answer to Exercise 3 (on page 11)

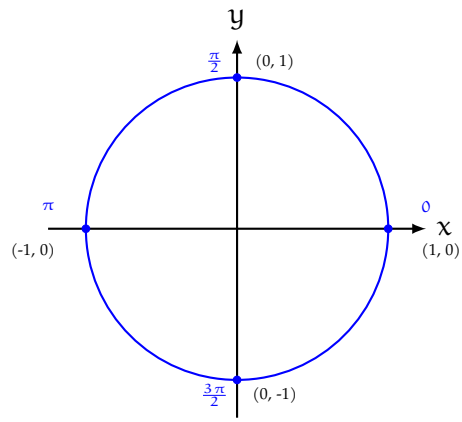
1. $\sec x + C$

Answer to Exercise 4 (on page 14)

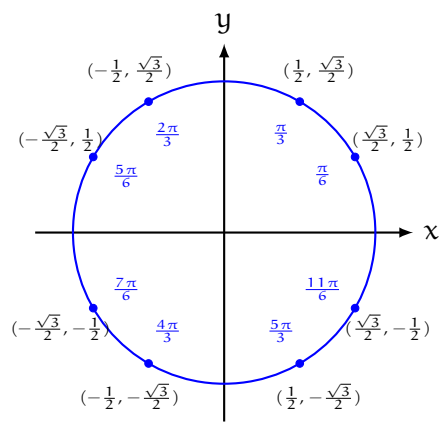
We know that for a right triangle, $\cos\theta = \frac{\text{adjacent leg}}{\text{hypotenuse}}$. For a right triangle inscribed in the Unit Circle, the adjacent leg is parallel to the x-axis and has the same length as the x-value of the coordinate point on the circle. Additionally, the length of the hypotenuse is 1. Therefore, $\cos\theta = \frac{x_o}{1} = x_o$.

Answer to Exercise 5 (on page 15)

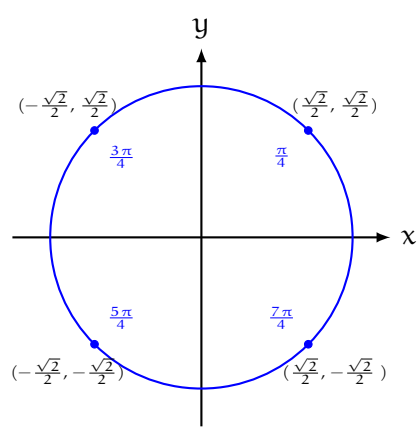
1. $\sin\frac{\pi}{2} = 1$
2. $\cos\frac{3\pi}{2} = 0$
3. $\sin\pi = 0$
4. $\cos -\pi = -1$ (Negative angles are measured clockwise from the x-axis, so $\theta = -\pi$ is at the same angle as $\theta = \pi$.)



Answer to Exercise 6 (on page 17)



Answer to Exercise 7 (on page 18)



Answer to Exercise 8 (on page 19)

1. 0
2. $\sqrt{2}/2$
3. $-1/2$
4. $-1/2$
5. $\sqrt{2}/2$
6. $1/2$
7. $\sqrt{2}/2$
8. -1
9. $-\sqrt{3}/2$
10. $1/2$

Answer to Exercise 9 (on page 21)

1. $\sin(\pi/12) = \sin(\pi/3 - \pi/4) = \sin\frac{\pi}{3}\cos\frac{\pi}{4} - \cos\frac{\pi}{3}\sin\frac{\pi}{4} = \left(\frac{\sqrt{3}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{1}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{\sqrt{6}-\sqrt{2}}{4}$
2. $\cos(7\pi/12) = \cos(4\pi/12 + 3\pi/12) = \cos(\pi/3 + \pi/4) = \cos\frac{\pi}{3}\cos\frac{\pi}{4} - \sin\frac{\pi}{3}\sin\frac{\pi}{4} = \left(\frac{1}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} = \frac{\sqrt{2}-\sqrt{6}}{4}$
3. $\tan(13\pi/12) = \frac{\sin(13\pi/12)}{\cos(13\pi/12)}$ First, we will find $\sin(13\pi/12)$: $\sin(13\pi/12) = \sin(3\pi/12 + 10\pi/12) = \sin(\pi/4 + 5\pi/6) = \sin\frac{\pi}{4}\cos\frac{5\pi}{6} + \cos\frac{\pi}{4}\sin\frac{5\pi}{6} = \left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{-\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{1}{2}\right) = \frac{-\sqrt{6}}{4} = \frac{\sqrt{2}}{4} = \frac{\sqrt{2}-\sqrt{6}}{4}$. Next we find $\cos(13\pi/12) = \cos(\pi/4 + 5\pi/6) = \cos\frac{\pi}{4}\cos\frac{5\pi}{6} - \sin\frac{\pi}{4}\sin\frac{5\pi}{6} = \left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{-\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{1}{2}\right) = \frac{-\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{-\sqrt{6}-\sqrt{2}}{4}$. And therefore $\tan(13\pi/12) = \frac{\sin 13\pi/12}{\cos 13\pi/12} = \frac{\sqrt{2}-\sqrt{6}}{4}\cdot\frac{4}{-\sqrt{6}-\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}+\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}+\sqrt{2}}\cdot\frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}-\sqrt{2}} = \frac{6-2\sqrt{12}+2}{6-2} = \frac{8-4\sqrt{3}}{4} = 2 - \sqrt{3}$

Answer to Exercise 10 (on page 22)

$\sin 2\theta = \sin\theta + \theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$

Answer to Exercise 11 (on page 23)

Similar to $\cos(\alpha/2)$, we begin with the double angle formula for cosine, but another version:

$$\cos 2\theta = 1 - 2\sin^2\theta$$

Substituting $\theta = \alpha/2$:

$$\cos\alpha = 1 - 2\sin^2(\alpha/2)$$

And rearranging to solve for $\sin(\alpha/2)$:

$$\sin(\alpha/2) = \pm\sqrt{\frac{1-\cos\alpha}{2}}$$

Answer to Exercise 12 (on page 27)

The volume of the sphere (in cubic meters) is

$$\frac{4}{3}\pi(1.5)^3 = 4.5\pi \approx 14.14$$

The mass (in kg) is $14.14 \times 7800 \approx 110,269$

The kinetic energy (in joules) is

$$k = \frac{110269 \times 5^2}{2} = 1,378,373$$

About 1.4 million joules.

Answer to Exercise 13 (on page 28)

In your mind, you can disassemble the tablet into a sphere (made up of the two ends) and a cylinder (between the two ends).

The volume of the sphere (in cubic millimeters) is

$$\frac{4}{3}\pi(2)^3 = \frac{32}{3}\pi \approx 33.5$$

Thus, the cylinder part has to be $90 - 33.5 = 56.5$ cubic mm. The cylinder part has a radius of 2 mm. If the length of the cylinder part is x, then

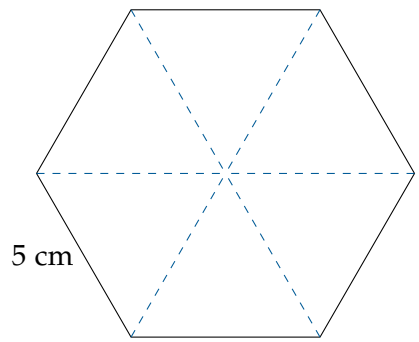
$$\pi 2^2 x = 56.5$$

Thus $x = \frac{56.5}{4\pi} \approx 4.5$ mm.

The cylinder part of the table needs to be 4.5mm. Thus the entire tablet is 8.5mm long.

Answer to Exercise 14 (on page 31)

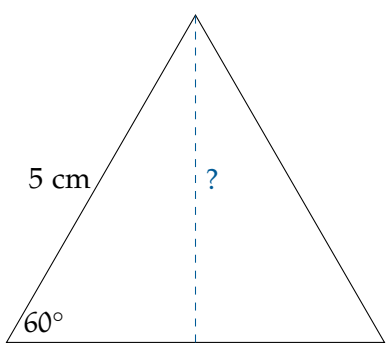
First, you need to find the area of the base, which is a regular hexagon:



All the angles in this picture are 60° or $\frac{\pi}{3}$ radians. This means each line is 5 cm long.

This tells us that we need to find the area of one of these triangles and multiply that by six.

Every triangle has a base of 5cm. How tall are they?



$$5\sin 60^\circ = 5\frac{\sqrt{3}}{2}$$

Which is about 4.33 cm.

So, the area of single triangle is

$$\frac{1}{2}(5)\left(5\frac{\sqrt{3}}{2}\right) = 25\frac{\sqrt{3}}{4}$$

And the area of the whole hexagon is six times that:

$$75\frac{\sqrt{3}}{2}$$

Thus, the volume of the pyramid is:

$$\frac{1}{3}hb = \frac{1}{3}13\left(75\frac{\sqrt{3}}{2}\right)$$

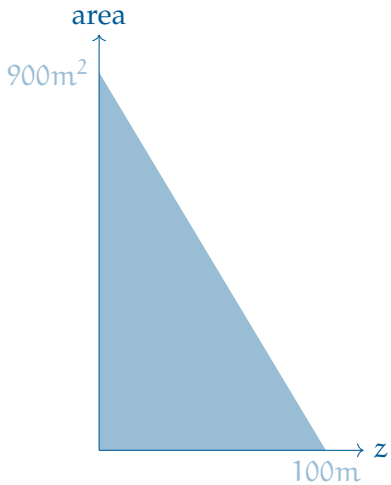
About 281.46 cubic centimeters.

Answer to Exercise 15 (on page 33)

The area at height z is given by:

$$a = \frac{1}{2}w^2 = \frac{1}{2}\left(30\sqrt{1-\frac{z}{100}}\right)^2 = \frac{1}{2}900\left(1-\frac{z}{100}\right)$$

If we plot that, it looks like this:



What is the area of the blue region? $\frac{1}{2}(900)(100) = 45,000$

The building will be 45 thousand cubic meters.

Answer to Exercise 16 (on page 42)

- $[1, 2, 3] + [4, 5, 6] = [5, 7, 9]$
- $[-1, -2, -3, -4] + [4, 5, 6, 7] = [3, 3, 3, 3]$
- $[\pi, 0, 0] + [0, \pi, 0] + [0, 0, \pi] = [\pi, \pi, \pi]$

Answer to Exercise 17 (on page 42)

To get the net force, you add the two forces:

$$F = [4.2, 5.6, 9.0] + [-100.2, 30.2, -9.0] = [-96, 35.8, 0.0] \text{ newtons}$$

Answer to Exercise 18 (on page 43)

- $2 \times [1, 2, 3] = [2, 4, 6]$
- $[-1, -2, -3, -4] \times -3 = [3, 6, 9, 12]$
- $\pi[\pi, 2\pi, 3\pi] = \pi^2, 2\pi^2, 3\pi^2]$

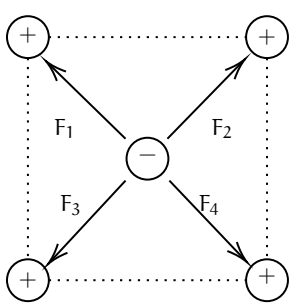
Answer to Exercise 19 (on page 48)

- $|[1,1,1]| = \sqrt{3} \approx 1.73$
- $|[-5,-5,-5]| = |-5 \times [1,1,1]| = 5\sqrt{3} \approx 8.66$
- $|[3,4,5] + [-2,-3,-4]| = |[1,1,1]| = \sqrt{3} \approx 1.73$

Answer to Exercise 20 (on page 52)

$$F = K\frac{|q_1 q_2|}{r^2} = (8.988\times10^9)\frac{(-5\times10^{-9})(-5\times10^{-9})}{0.12^2} = \frac{224.7\times10^{-9}}{0.0144} = 15.6\times10^{-6}\,\text{N or}15.6\ \text{micronewtons}$$

Answer to Exercise 21 (on page 54)



The attractive forces of F_1 and F_3 cancel each other out. Similarly, the attractive forces of F_2 and F_4 also cancel each other out. The net charge on the negative point charge is zero.

Answer to Exercise 22 (on page 56)

We noted above that the electric field caused by a point charge is defined as $E = \frac{kq}{r^2}$. We need to find the electric fields for each charge and then subtract their strengths.

Inputting known variables as constants for X from point P:

$$E = \frac{kQ}{d^2}$$

And doing the same for Y:

$$E = \frac{2kQ}{(2d)^2} = \frac{kQ}{2d^2}$$

Because the fields are in opposite directions, we subtract the smaller magnitude from the larger one. Let's define *to the right* as the positive direction:

$$E_{net} = E_X - E_Y \implies \frac{kQ}{d^2} - \frac{kQ}{2d^2}$$

To subtract, we find a common denominator:

$$E_{net} = \frac{2kQ}{2d^2} - \frac{kQ}{2d^2} = \frac{kQ}{2d^2}$$

The correct answer is C.

Answer to Exercise 23 (on page 61)

The amount of charge dQ on a small *surface* section of the belt is $dq = \sigma\, dA$. For a belt of width w, the area passing the wire in time dt is $dA = w(v\, dt)$. Substituting this in:

$$I = \frac{\sigma(w \cdot v \cdot dt)}{dt} = \sigma \cdot w \cdot v \tag{6.12}$$

Plugging in our values:

$$I = (5.0 \times 10^{-3}\ \mathrm{C/m^2}) \cdot (0.60\ \mathrm{m}) \cdot (2.5\ \mathrm{m/s})$$
$$I = 7.5 \times 10^{-3}\ \mathrm{A} = 7.5\ \mathrm{mA}$$

Answer to Exercise 24 (on page 62)

Since the field is uniform, the charge has a constant electric force felt on it: $\vec{\mathbf{F}}_E = q\vec{\mathbf{E}}$. Since q is positive, $\vec{\mathbf{F}}_E$ points in the same direction as $\vec{\mathbf{E}}$. Because the force and the displacement r are in the same direction, the work W_E done by the electric field is positive:

$$W_E = F_E \cdot r = qEr$$

Thus the change in electric potential energy is

$$\Delta U_E = -qEr$$

It is *negative* because a doing positive work results in negative change in potential energy.

Answer to Exercise 25 (on page 63)

Using our equations, we can derive an equation for the potential difference V:

$$\begin{aligned}
\Delta U_E &= q\Delta V \\
W_E &= \Delta U_E \\
W_E &= qV \\
\text{gain in kinetic energy} &= \text{work done by field} \\
\frac{1}{2}mv^2 &= qV \\
V &= \frac{1}{2}\frac{mv^2}{q}
\end{aligned}$$

Since our we are given the speed, and the charge and mass of the electron, we can plug in our values:

$$\begin{aligned}
V &= \frac{\frac{1}{2}(9.11 \times 10^{-31})(9.4 \times 10^{6})^2}{1.60 \times 10^{-19}} \\
&= \frac{\frac{1}{2}(9.11 \times 10^{-31})(8.836 \times 10^{13})}{1.60 \times 10^{-19}} \\
&= \frac{\frac{1}{2}(8.05 \times 10^{-17})}{1.60 \times 10^{-19}} \\
&= \frac{\frac{1}{2}(8.05 \times 10^{-17})}{1.60 \times 10^{-19}} \\
&= \frac{4.03 \times 10^{-17}}{1.60 \times 10^{-19}} \\
&= 252\,\mathrm{V}
\end{aligned}$$

So each coloumb of charge is accelerated by a potential difference of 252 volts. And since the electron has a charge of 1.6×10^{-19} C, the total energy gained by the electron is all transferred into kinetic energy.

Answer to Exercise 26 (on page 65)

First off, why do we know that the electron will hit a plate? Because the electric field between the plate exerts a force on the electron. The electron is attracted to the **positively** charged upper plate, so it will accelerate upwards and eventually hit the upper plate. Let's first find. the acceleration caused by the electric field. The electric field strength E

between two parallel plates is given by:

$$E = \frac{V}{d} = \frac{30}{0.040} = 750\,\mathrm{V\ m^{-1}} \tag{6.20}$$

Converting this to solve for the alectron's acceleration:

$$a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.6 \times 10^{-19})(750)}{9.11 \times 10^{-31}} \approx 1.32 \times 10^{14}\,\mathrm{m\ s^{-2}} \tag{6.21}$$

To find the horizontal distance travelled, we use the vertical motion to find time, and then use that time to find the horizontal distance travelled. The vertical motion is a simple kinematics problem with constant acceleration:

$$\begin{aligned} y &= v_{y0}t + \frac{1}{2}a_y t^2 \\ 0.02 &= 0 + \frac{1}{2}(1.32 \times 10^{14})t^2 \\ t^2 &= \frac{0.04}{1.32 \times 10^{14}} \\ t &\approx 1.74 \times 10^{-8}\,\mathrm{s} \end{aligned}$$

Then, putting this into a horizontal motion equation:

$$\begin{aligned} x &= v_{x0}t \\ &= (9.4 \times 10^{6})(1.74 \times 10^{-8}) \\ &\approx 0.16\,\mathrm{m} \end{aligned}$$

So the electron travels a horizontal distance of about **0.16 meters** before hitting the upper plate.

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